SafeCircle: Super Master Document (v2)

Oct 1, 2026 · @iCodejr

0. How to use this document

This is version 2 and the single source of truth for SafeCircle. It merges every line of v1 with the new two-faced design: a phone-free person side (Face A) and a volunteer app (Face B).

v2.1 update (1 Oct 2026): the button is now a family of two devices, a Wi-Fi **Home button** and a **Go pendant** with GPS and 4G, so the person can be found away from home. Section 18 is rewritten as "Hardware and location"; Sections 1-10, 15-17 and 19-27 are updated to match.

v2.2 update (1 Oct 2026): new **Stealth mode** for girls and women (Section 7.2). The button confirms each stage with light only, with no beeps and no calls to her phone, so people nearby cannot tell an alert was raised while help is on the way. Sections 5, 6, 7, 10, 16, 17, 21, 25 and 26 are updated to match.

To build a deck, paste or upload this doc and say: *"Using this document, make an [N]-slide deck for [audience]. Focus on [sections]."* A suggested deck outline is in Section 25.

What changed from v1

Area	v1	v2
Product shape	Button and WhatsApp for the person; volunteer web app	Two faces: Face A (button, WhatsApp/SMS, setup web portal) and Face B (native volunteer app)
Cancel	Press again inside a 3-5 s window	Long-press cancel at any time, then a verification call. No answer means the alert is real
Button feedback	Sending, sent, help coming	Full pattern table: sending, received, accepted, arriving, arrived, escalated, failed, verifying, plus a discreet mode
Alerting	Volunteers within about 500 m, first to accept	Tiered: nearest 3 verified get high priority, everyone else a normal alert, all upgraded at 30 s

Area	v1	v2
Volunteer location	Registered address, fresh reading on accept	Four battery-aware layers: zone, hourly refresh, wake-up push on alert, live tracking only while en route
Trust	Basic verification step	Government ID, occupation, verified tag, society verifier and community vote
Training	Optional course	Mandatory app tutorial plus an emergency tutorial library
Responder aid	First-aid cards	Smart flashcards ranked from the person's profile and history, tap to confirm
Scenarios	Demo stories	A playbook of real-world scenarios and edge cases (Section 16)
Locating the person	Registered address; outdoors only by asking for a WhatsApp location	Go pendant with GPS, Wi-Fi scan, cell and 4G: best location within seconds, then live updates during the alert (Section 18)

Labels used

- **(est.)** an estimate from memory. Re-check before presenting as fact.
- **Roadmap** not part of the prototype. Never present it as working.
- **Verify** checked against public sources on 1 Oct 2026, but rules change. Re-check before building or quoting.
- **Decision needed** listed again in Section 26.

The cyclone-detection deck used in v1 was a style reference only. Do not reuse its logos or branding.

1. Project snapshot and the problem

SafeCircle is a neighbourhood safety network that works even when the person in trouble cannot use a smartphone. It is the "first five minutes" layer that extends 112, not a replacement for it.

Theme: Connected Communities. **Working name:** SafeCircle (placeholder; not checked for existing use or trademarks).

Problem statement (as given):

> Accelerating Emergency Response and Enhancing Safety Accessibility Across Communities. During critical safety incidents and localized emergencies, residents in residential neighborhoods and public spaces face severe friction in initiating timely calls for help and obtaining actionable situational updates. Traditional reporting channels are often fragmented, causing delays in dispatching local authorities, coordinating nearby volunteers, and providing clear status updates to those affected. The issue is further exacerbated for vulnerable demographics, such as children, senior citizens, and individuals with physical or cognitive disabilities, who may find navigating complex smartphone applications difficult or impossible during high-stress scenarios. When an emergency strikes, relying exclusively on app-based reporting excludes these groups from receiving rapid, accessible assistance.

The idea in one paragraph: a child, senior or person with a disability presses one physical button, or a family member sends "HELP" on WhatsApp. Within seconds the system knows who they are and where they are: the registered address at home, or the Go pendant's GPS outside. It alerts the nearest verified volunteers, the society guard and the family at the same time, and escalates to 112 if nobody responds. The button tells the person, by light, sound and vibration, that help is coming. Volunteers arrive already knowing who they are helping and how.

One-line pitch: *"112 gets the police there. We make sure the person is never alone, and never phone-dependent, in the minutes before they arrive."*

Deck line: *"No app for victims. A responder app for volunteers."*

The broken chain the PS describes

Link	Where it breaks today	SafeCircle fix
1. Trigger	Too many steps for a person in panic, pain or confusion	One press, or one WhatsApp word, from a registered number
2. Routing	Police, ambulance, guard and neighbours are reached separately	One incident fans out to everyone in parallel
3. Coordination	Nearby people who could arrive first are never asked	Tiered, geofenced volunteer alerts with an accept timer
4. Feedback	Person and family hear nothing	Status on the button itself and live updates to family

Hidden requirement: the last paragraph of the PS. An app-only answer fails it by definition, so at least one non-app, low-effort trigger is mandatory.

Success criteria used throughout

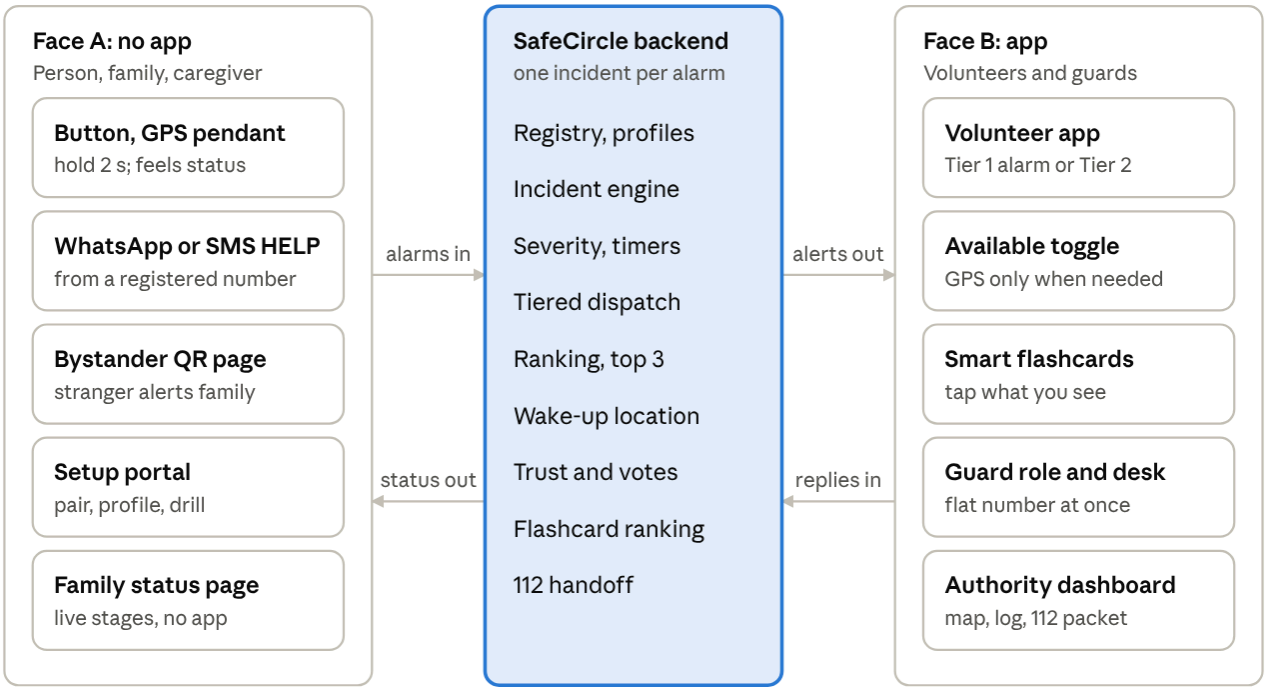
- **C1** Faster initiation of help
- **C2** Unified reporting and dispatch to authorities
- **C3** Coordinating nearby volunteers
- **C4** Clear status updates to those affected
- **C5** Accessible to children, seniors and people with disabilities without depending on an app

A strong answer needs more than one way to raise an alarm, one backend, a volunteer layer, a status loop back to the person, accessibility by design, and realistic handling of false alarms, privacy and poor connectivity.

2. The two-faced solution at a glance

SafeCircle has two faces joined by one backend. Face A serves the person in trouble and their family with no app at all; Face B is an app for the people who respond.

Two faces share one backend: no app for the person, an app for responders



two faces, one backend · 10 parts

Face A only sends alarms and receives status. Every decision about who to alert is made in the backend, and Face B receives alerts and sends back replies.

	Face A: person and caregiver	Face B: responders
Who	Child, senior or person with a disability; their family or caregiver	Volunteers (verified and unverified), society guards
Tools	Home button (Wi-Fi) and Go pendant (GPS, 4G), WhatsApp/SMS, bystander QR, setup web portal, family status page	Native volunteer app (web app for the demo), guard role, dashboard
Main jobs	Raise an alarm in one action; feel that help is coming; set up the profile once	Get the right alert, decide in seconds, reach the person, help well
Never needs	A smartphone app, reading, speaking or hearing	Constant GPS (battery-aware location)
Sections	4, 5, 6, 7	9, 10, 11, 12, 13, 14

2.1 Actors

Actor	Role	Sees
The person	Presses the button or sends "HELP"	Light, sound and vibration on the button; WhatsApp replies
Caregiver or family	Sets up the profile; can trigger, confirm or follow an alert	Live status page; verification messages
Verified volunteer	Tier 1 responder; votes on new members	Category and distance first; full briefing after accepting
Unverified volunteer	Tier 2 responder with limits	The same, but never sole responder for protected profiles
Society guard	On-site responder; opens gates; in-person checks	Full address and access notes immediately
Society verifier or RWA admin	Approves local volunteers; bulk onboarding	Volunteer cards for approval; monthly report
Authority (112, police)	Takes over when needed	Dashboard and handoff packet

Actor	Role	Sees
SafeCircle admin	Reviews concerns, misconduct, content	Audit logs and reports

2.2 End-to-end flow

1. **Setup (once):** caregiver pairs the button, fills the profile and runs a drill (Section 5).
2. **Trigger:** hold the button, send "HELP", or a stranger scans the pendant QR (Section 4).
3. **Identify and score:** the backend finds the person, locates them (registered address, or GPS and Wi-Fi from the pendant), merges duplicates and sets severity (Section 8).
4. **Alert in tiers:** 3 nearest verified volunteers get a high-priority alarm, everyone else a normal alert, guard and family at the same time; all upgraded at 30 s; 112 at 90 s or at once for P1 (Section 9).
5. **Accept and go:** fresh location, full briefing, smart flashcards, live status taps (Sections 10, 11, 14).
6. **Feedback:** the button shows each stage; the family page updates (Sections 7, 15).
7. **Cancel path:** a long-press cancel triggers a verification call; no answer keeps the alert alive (Section 6).
8. **Close and learn:** outcome logged, profile history updated, volunteer wellbeing check (Sections 8, 10).

2.3 Core differentiators

1. **Phone-free trigger** for people who cannot use an app, at home or outside (GPS pendant).
2. **Tiered, verified-first alerting** that reaches everyone at once but wakes the closest trusted people hardest.
3. **Cancel with verification:** a forced cancel does not end help.
4. **Status the person can feel:** light, sound and vibration for each stage.
5. **Profile-aware dispatch and smart flashcards:** responders know who they are helping and what they will probably find.
6. **Community-vouched trust:** neighbours who know a volunteer approve them.
7. **Battery-aware responder location:** exact position only when someone nearby needs help.

3. Research: existing systems, prior art and gaps

India's 112 system already accepts IoT and external signals, and proven volunteer-alert apps abroad use the same model we propose. That makes SafeCircle a realistic "first five minutes" layer that can later plug into 112, not a fantasy.

3.1 India: 112 / ERSS (national)

- The MHA ERSS page says ERSS-112 is state-run and receives distress signals over 10 channels: voice call, SMS, SOS, email, web request, chatbot, media crawler, IoT-based signals, WhatsApp and external signals. Calls are passed to police, health, fire, disaster, women, children or railway dispatchers on GIS-based dispatch terminals.
- **Coverage conflict from v1 resolved.** A Rajya Sabha answer (Dec 2023) says ERSS is operational in all 36 states and UTs. The "24" figure is the ERSS 2.0 upgrade: 24 states and UTs, including Tamil Nadu and Puducherry, had moved to ERSS 2.0, with the rest expected by March 2026. **Verify** current ERSS 2.0 status before quoting.
- **112 India app:** a button sends an alert with location and calls 112 (112.gov.in). Its SHOUT feature enlists volunteers registered with the state police, aimed mainly at women (OneIndia).
- **Phone triggers (from v1):** dial 112; press the power button 3 times; long-press 5 or 9 on a feature phone.
- **Why it matters for us:** "IoT-based signals" and "external signals" are already listed channels. Real integration is still Roadmap and needs permission, but the door exists.

3.2 Tamil Nadu and Chennai (from v1, not re-checked)

- **Kavalan SOS** (TN Police, June 2018): red SOS button, 5-second countdown, sends location and back-camera video to police; callback within about a minute; SMS to registered contacts. App-based and police-only.
- **Kaaval Karangal** (Greater Chennai Police, April 2021): helpline and rescue team for abandoned elderly people. A service you call, not a trigger.

3.3 Prior art abroad: volunteer alert systems that work

System	What it does	Lessons we copy
GoodSAM (UK and others)	Alerts verified responders near a suspected cardiac arrest; integrated with most UK ambulance dispatch systems	Verifying organisations check ID and certificates; responders must switch on "Report on Duty"; SMS alerts where push is not possible

System	What it does	Lessons we copy
GoodSAM in Northern Ireland (NIAS FAQ)	Alerts only within 500 m, responder walks; siren sound; accept or reject; alert disappears after 15 s	Location accuracy slider: cell triangulation is described as under 1% battery, GPS as power hungry; GPS stops on arrival; "stand down" messages; mission log after each call; no alerts for assaults or overdoses; certificates expire and alerts stop
GoodSAM in New Zealand (St John)	1,000 m alert radius; shows nearest AEDs	Larger radius where volunteers are sparse
SMS Lifesavers, Stockholm (Karolinska)	Randomised trial: about 10,000 CPR-trained volunteers alerted by text within 500 m	Bystander CPR before ambulance rose to 62% vs 48% in the control group (CBS News). Hard evidence that geofenced volunteers arrive first

GoodSAM's onboarding also asks new responders to read a code of conduct and run a simulation before their first alert (St John poster). Our mandatory tutorial (Section 13) follows the same idea.

3.4 Gaps SafeCircle fills

1. **The trigger still needs a smartphone.** 112 India, the power-button shortcut and Kavalan all assume a working phone in hand.
2. **Everything is police- or ambulance-centric.** Guards, RWAs, neighbours and nearby doctors sit outside the loop.
3. **Volunteer layers are narrow.** SHOUT targets women; GoodSAM targets cardiac arrest only.
4. **No status loop to the person.** Nobody tells a frightened senior, by touch or light, that help is coming.
5. **No building-level coverage.** Nothing links to society security, gates or flats.
6. **No profile-aware briefing.** Every caller is treated the same.
7. **No cancel verification.** No system we found checks whether a cancel was made under duress.

3.5 Comparison (deck table)

	112 India / Kavalan	GoodSAM	SafeCircle
Trigger	Smartphone app or call	999 call or Alerter app	Phone-free button, WhatsApp/SMS, caregiver, bystander QR
Routing	Control room to police, fire, health	Ambulance dispatch to responders	One incident to volunteers, guard, family and 112 in parallel
Volunteers	SHOUT, police-registered	Verified, cardiac arrest only	Verified and community-vouched, all emergency types, tiered by distance
Feedback to person	Callback, SMS to contacts	None to patient	Light, sound and vibration on the button; live family status
Responder briefing	Not found	Location and AEDs	Profile, handling notes, smart flashcards
Cancel safety	Countdown only	n/a	Long-press cancel plus verification call
Locating the person	Phone GPS, only if the person can use the phone	Location given by the 999 caller	Registered address at home; GPS, Wi-Fi scan and cell from the Go pendant outside

3.6 Still not researched

Phone-maker features (Android Emergency SOS, Apple Emergency SOS) and paid family-safety apps exist. Check them before claiming nothing similar exists.

4. Face A: how the person raises an alarm

Face A gives the person five ways to call for help, and none of them needs an app. Every trigger becomes the same incident event in one backend (Section 8), so routing never depends on the channel.

#	Trigger	Who it suits	Effort for the person	Tier
T1	Home button (wall, bedside) or Go pendant with GPS and 4G	Children, seniors, cognitive and physical disabilities	One press	**Must**
T2	WhatsApp "HELP" (or any message) from a registered number	Anyone with a basic smartphone	One message, or one tap on a fridge QR that opens a prefilled chat	**Must**
T3	Caregiver trigger (WhatsApp, web link or call on behalf of a dependent)	Severe cognitive disability; worried family	One message	Should
T4	SMS keyword or missed call to a SafeCircle number	Feature-phone users	One SMS, or ring and hang up	Roadmap
T5	Bystander QR on the pendant (scan, tap "Alert family", share location)	Lost child or wandering senior found by a stranger	Done by the stranger	Should

T1. Physical button (hero feature)

- No screen, login or reading. A long press (about 2 s) starts the alert, so a brush does not.
- Two devices (Section 18): the Home button, Wi-Fi only, fixed at bed, bathroom or wheelchair height with a large palm or elbow surface; and the Go pendant, with GPS and 4G, worn as a pendant, wristband, bag tag or school-ID holder outside the home.
- Sends its device ID and event, plus a location from the Go pendant. Name, address and medical notes live on the server.
- Full behaviour (cancel, verification) is in Section 6; feedback patterns in Section 7; hardware in Section 18.

T2. WhatsApp "HELP"

- Register first. A message carries only the sender's number, so the registry turns that number into a person, an address and a profile.

- The provider pushes each message to our webhook. The backend verifies the signature, normalises it to `{source: "whatsapp", from: "+91...", text: "HELP"}`, looks up the number, then runs the same pipeline as the button.
- Location: the registered address by default. Outdoors, the reply carries a "Share location" request.
- Pricing (**Verify**): since 1 July 2025 Meta charges per template message. When a user messages us, a 24-hour customer service window opens; free-form replies and utility templates inside it are free (Meta pricing docs). Status replies to the person are therefore free. Alerts we start ourselves to volunteers or family outside a window need paid, pre-approved utility templates. This is one reason volunteers get a native app.
- Unregistered numbers still get help, as an **unverified** incident: auto-reply asking for location and one line of detail, alert to the security desk and dashboard, 112 escalation shown. A stranger's location is not auto-sent to volunteers (abuse risk; **Decision needed**).
- Abuse controls: "CANCEL" reply window (then the verification call in Section 6), merge repeat messages within a few minutes, per-number rate limit.

T3. Caregiver trigger

A caregiver's number is registered against one or more dependents. "HELP" from it raises an alarm for the dependent. With several dependents, the bot replies with a short list, or accepts "HELP Lakshmi". The incident is marked "raised by caregiver, location = dependent's address" so responders do not go to the caregiver's location.

T4. SMS keyword and missed call (Roadmap)

- Inbound SMS in India needs an Indian virtual-number provider; v1 notes Twilio SMS to India is outbound-only (**Verify**).
- A missed-call number is a familiar Indian pattern and free for the caller: ring and hang up, the caller ID is matched against the registry. Same pipeline. Provider and cost not yet researched.

T5. Bystander QR (new)

A small QR on the back of the pendant opens a web page with no personal data on it: "This person may need help. Tap to alert their family." The stranger can share their phone's location and type a note. The family and nearest guard or volunteers get an alert with that location. It is the backup for lost children and wandering seniors when the pendant has no GPS fix or no battery, or when the person only has a Home button. It must never show the person's name, address or medical notes to the scanner.

5. Face A: setup and onboarding web portal

A caregiver, family member or society admin sets everything up once on a mobile web page; the person in need never touches it. The portal pairs the button, connects it to Wi-Fi, records the profile and runs a test, in about 10 minutes (est.).

5.1 Onboarding steps

1. **Sign in** with the caregiver's phone number and a one-time code.
2. **Add the person** (profile fields in 5.2). One caregiver can manage several people; one person can have several caregivers.
3. **Pair the button.** Scan the QR on the device box. It carries the device ID and a one-time claim code, so a stolen or second-hand button cannot be claimed by someone else. A Go pendant also has its SIM activated here.
4. **Connect the button to Wi-Fi.** Two options, both browser-based:
   - **SoftAP captive portal:** the button opens its own hotspot; the phone joins it and a setup page pops up. Works in any browser, including iPhone.
   - **Improv Wi-Fi over Bluetooth:** an open standard that sends Wi-Fi credentials over BLE from a web page in Chrome or Edge, then redirects back to our portal to finish onboarding (Home Assistant). Not available in iPhone browsers, which lack Web Bluetooth.
   - Demo: credentials hardcoded to the phone hotspot (from v1). Several networks can be saved and named (home, school, day centre); the one the button joins tells the system where it is.
5. **Add trigger numbers** (`self`, `caregiver`), each confirmed by a one-time code.
6. **Add family contacts** in call order, and choose who may cancel an alert on the person's behalf.
7. **Choose the feedback style** for the button (Section 7): standard, loud, calm, discreet or stealth. For a Go pendant, also choose location sharing: during alerts and drills only (default), or opt-in safe-zone mode (Roadmap).
8. **Consent:** the person (or guardian for a minor or a person lacking capacity) agrees to what is stored and who sees it.
9. **Run a drill.** A test press goes through the whole chain marked `drill`: the button plays every pattern, the caregiver gets sample messages, and no volunteer is alerted.
10. **Print the kit:** fridge QR (prefilled WhatsApp "HELP"), wallet card with the SafeCircle number, and the bystander QR sticker for the pendant.

5.2 Profile fields

Every field must change how someone responds. Anything that does not is left out (data minimisation).

Group	Fields	Who sees it
Identity	Name, name they respond to, photo, age band, languages	Accepted responder, guard
Location	Address, block and flat, landmark, pin on map, society, where the key is kept, known places (school, day centre) and their Wi-Fi networks	Accepted responder, guard
Access secrets	Gate or door code, spare-key holder	Accepted guard only, encrypted
Health (optional)	Conditions, medicines, allergies, blood group, mobility aids, regular doctor and hospital	Accepted responder; summary to 112 handoff
Communication and handling	Hearing, vision and speech needs; calming steps; things to avoid ("do not touch", "no loud voices")	Accepted responder
Safety rules	Child: who may collect; people who must never get information (custody or abuse cases); wandering history	System rules plus accepted responder
Responder preference (optional)	For example a woman responder first	Dispatch logic only
Device	Feedback style, volume, vibration strength, LED brightness	Device and caregiver
Consent	Who consented, when, what for	Audit only

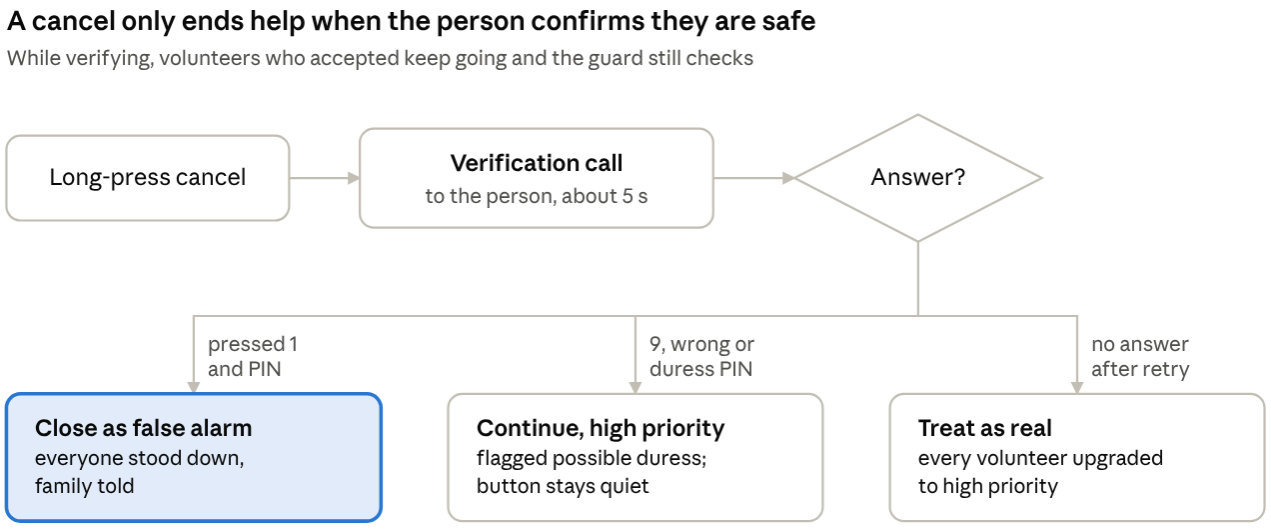
5.3 After setup

- **Caregiver home page:** button battery, last check-in, Wi-Fi, 4G and GPS status, incident history and a profile-completeness score.

- **Health checks:** heartbeat missing or battery low triggers a message to the caregiver (Section 8).
- **Reminders:** a monthly drill and a six-monthly profile review, because medicines and conditions change.
- **Society bulk onboarding:** the RWA or society admin can register many households and buttons from one sheet, then each family confirms its own details.

6. Button behaviour: press, cancel and verification call

A cancel never silently ends an alert. A long press cancels, the system then calls the person to confirm, and if nobody picks up the alert is treated as real and continues. This protects against the worst case: an attacker forcing the person to cancel.



cancel and verification flow · 1 decision, 3 outcomes

Only one of the three outcomes ends help. Silence, a wrong answer or the duress PIN keeps the alert alive.

6.1 Press rules

Action	Gesture	Result
Raise SOS	Hold about 2 s	Incident created at once; dispatch starts with no waiting
Request cancel	Hold about 4 s (longer than the trigger, so the two cannot be confused)	Incident moves to **Cancel requested, verifying**

Action	Gesture	Result
Silent SOS (discreet profiles only)	Triple short press	Alert with no sound or light on the button; no call to the person
Brush or tap	Under 2 s	Ignored

v1 used a 3-5 s countdown before sending. v2 sends immediately, because the cancel-and-verify path now handles mistakes. A pre-send countdown stays as a per-person option (0-5 s) for people who press by accident often. **Decision needed.**

6.2 What happens after a cancel

1. **Within about 5 s** the backend places an automated verification call (IVR) to the person's own registered phone. The caller ID shows SafeCircle.
2. **The call says**, in the person's language: "Your SafeCircle alert was cancelled. Press 1 if you are safe. Press 9 if you still need help." An optional 4-digit safe PIN can be asked after pressing 1.
3. **Outcomes:**
   - **Pressed 1 (and correct PIN if set):** incident closed as false alarm. Everyone gets "Stand down, person confirmed safe". The family is told.
   - **Pressed 9, wrong PIN, or the duress PIN:** the alert continues at **high priority**, flagged "possible duress". The button still shows the cancelled pattern so an attacker is not tipped off.
   - **No answer, busy or call rejected:** one retry after about 10 s. Still nothing means the alert is **real**. It resumes with every volunteer at high priority (Section 9).
4. **The caregiver is told in parallel** ("Lakshmi's alert was cancelled, we are checking"). A caregiver can confirm safety too, but for high-risk profiles a caregiver's "safe" does not stand down the guard's in-person check.

While verifying: volunteers who already accepted keep going ("Cancel requested, verifying, continue unless stood down"). No new volunteers are added. The guard still checks the flat, and a Go pendant keeps sending its location silently. Decision by about 60 s at most (est.).

6.3 When a call cannot work

Person	Verification instead
Child with no phone	Call the parent; guard or nearest verified volunteer does a doorstep or on-site check

Person	Verification instead
Deaf or speech-impaired	WhatsApp or SMS "Reply SAFE" plus the button's own confirm: the button asks with a vibration pattern, a double press means safe
Dementia or cognitive disability	A cancel from the person is never enough; caregiver confirms and the guard checks
Violence or stalking risk (discreet or stealth profile)	No call to the person at all (a ringing phone can endanger them); verification goes to the caregiver and responders continue

6.4 Optional check-in call at the start

For people with a phone and no discreet or stealth flag, the same IVR can call **in parallel** when an SOS starts: "Press 1 medical, 2 fall, 3 danger, 4 pressed by mistake." It never delays dispatch. No answer means nothing changes. "4" behaves like a cancel and goes through the verification above. The answer also seeds the volunteer flashcards (Section 14).

6.5 Provider

An Indian cloud-telephony provider such as Exotel offers outbound calls into an IVR flow with DTMF key collection, status webhooks to our server, and a Mumbai data centre (Exotel Voice v1 docs). It also lists Truecaller verified-business caller names and built-in DLT compliance (Exotel services), which helps the person trust the call. **Verify** pricing and KYC lead time. Demo: simulate the call with a dashboard button, or use a provider trial account.

7. Button feedback: what the person sees, hears and feels

The button reports every stage of the response through three senses at once, so a blind user, a deaf user and a child can each follow it. Each stage has its own colour, sound and vibration count, so no single sense is required.

7.1 Pattern table (standard profile)

Stage	Light (RGB LED)	Sound	Vibration	Meaning for the person
Holding	White ring fills over 2 s	Rising ticks	Light buzz	"Keep holding"

Stage	Light (RGB LED)	Sound	Vibration	Meaning for the person
Sending	Blue fast blink	Short beep each second	1 short	"Sending your call for help"
Received	Blue solid 2 s, then slow blue pulse	Two-tone chime	1 long	"We got it"
Accepted	Green slow pulse	Three rising notes ("help is coming")	2 short	"Someone is on the way"
Arriving (responder under about 100 m)	Green fast pulse	Same chime, faster	3 short	"They are almost here"
Arrived	Green solid	One long soft tone	1 long	"Help is with you"
Escalated to 112	Purple slow pulse	Distinct two-note chime	2 long	"Police or ambulance are also coming"
Still working (reassurance)	Current colour brightens	None	1 gentle buzz every 30-60 s	"Your alert is still active"
Cancel requested or verifying	One white flash, then off	None	None	Deliberately quiet (Section 6)
Failed after retries	Red fast blink	Triple beep	Long-long-long	"Not sent, still trying"
Low battery (idle)	Red blink every 30 s	None	None	Caregiver is also told

7.2 Feedback profiles (set in the portal)

Profile	For	Changes
Standard	Most people	As in 7.1
Loud	Seniors with hearing loss	Maximum buzzer volume, strongest vibration, repeated chimes

Profile	For	Changes
Calm	Autism, dementia, anxiety	Soft tones only, slow fades, no beeping; avoids sensory overload
Child	Children	Bright friendly colours, gentle chime, nothing alarming
Visual-first	Deaf users	Brighter LED, stronger vibration, sound optional
Touch-and-sound	Blind users	Distinct vibration counts and tones; colour irrelevant
Discreet	Harassment, stalking, domestic risk	Vibration only, no light or sound, no calls to the person
Stealth	Girls and women who may be in danger in public (harassment, being followed)	Lights only: the light patterns from 7.1 at low brightness, on the side facing her; no buzzer, no vibration, no calls to her phone; responders are told to arrive quietly

A voice prompt in Tamil, Hindi or English ("Help is coming, about 2 minutes away") is Roadmap. It needs a small speaker and audio module.

7.3 How the button learns each stage

A plain HTTP POST is one-way, so the button must stay in touch after a press:

- **Demo:** after a press, the button keeps Wi-Fi on and polls `GET /api/device/{id}/status` every 3 s for 5 minutes, then every 15 s until the incident closes (timings est.).
- **Better:** MQTT. The button subscribes to its own topic and gets each stage pushed instantly.
- Each status carries a sequence number, so a late or repeated message never moves the button backwards. The Go pendant receives status over 4G, so feedback also works away from home.
- Polling or a live connection drains the battery. That is acceptable during an emergency; between emergencies the button sleeps and only wakes for a heartbeat (Section 18).

8. Incident engine: severity, states, timers and escalation

Every trigger becomes one incident with a severity, a state and a set of running timers. Severity decides whether 112 is told at once or only after volunteers fail to respond.

8.1 Pipeline

1. **Receive** an event from any channel (button, WhatsApp, SMS, missed call, caregiver link, bystander QR). Verify its signature or token.
2. **Normalise** to `{source, sender_or_device, type, timestamp}`.
3. **Identify** the person from the registry and locate them: registered address, known place, or the pendant's GPS, Wi-Fi scan or cell position, whichever is most accurate (Section 18). No match means an **unverified** incident (Section 4, T2).
4. **Merge** duplicates: the same person within 10 minutes joins the open incident instead of creating a new one.
5. **Score severity** (8.2).
6. **Dispatch** in parallel: tiered volunteers (Section 9), guard, family, and 112 if P1.
7. **Track** states and timers (8.3, 8.4) until a responder or the dashboard closes it.

8.2 Severity levels

Rules first; AI triage only suggests (Should tier).

Level	Typical inputs	Dispatch
P1 Critical	Cardiac or seizure history with no answer to the check-in call; caregiver reports unconscious; check-in "3 danger"; possible duress after a cancel; a child or dementia profile whose pendant moves at vehicle speed (possible abduction; Decision needed)	Volunteers all at high priority, guard, family, and 112/108 told **at once** (simulated in the demo)
P2 Urgent (default for a button press)	Any press or "HELP" from a registered person	Tiered volunteers, guard and family; 112 if nobody accepts in 90 s
P3 Assist (Roadmap mode)	A "need help, not an emergency" press, for example a senior stuck after a minor fall	Volunteers and family; no automatic 112
Unverified	"HELP" from an unknown number	Guard desk and dashboard; location requested; 112 shown

8.3 Incident states

State	Entered when	Leaves to
New	Event identified	Dispatching
Dispatching	Alerts sent	Accepted, Escalated, Cancel verifying
Accepted	First responder accepts	En route
En route	Responder taps "On my way"	On scene
On scene	Responder taps "Arrived" or is within about 30 m	Resolved, Handed to 112
Cancel verifying	Long-press cancel or "CANCEL"	Closed as false alarm, or back to Dispatching at high priority
Escalated (a flag, not an end)	112 told	Continues alongside the others
Resolved / Closed	Responder, guard or dashboard closes with an outcome	Archive and follow-up

An incident is never auto-closed while unresolved. A stale one is flagged on the dashboard instead.

8.4 Timers (P2 default; all est., tune in the pilot)

Time	Check	Action if it fails
0 s	Alert sent	Top 3 high priority, everyone else normal (Section 9)
30 s	Did any of the top 3 accept?	Upgrade every alerted volunteer to high priority; widen the radius
60 s	Has the family opened the message?	Automated call to the family, then the next contact
90 s	Has anyone accepted?	Escalate to 112 (simulated); guard and society admin called
Accept + 3 min	Is the responder moving closer?	Ask "Still coming?"; if no reply, re-open to others

Time	Check	Action if it fails
On scene + 30 min	Any update?	Check-in ping to responder and family

8.5 How many responders

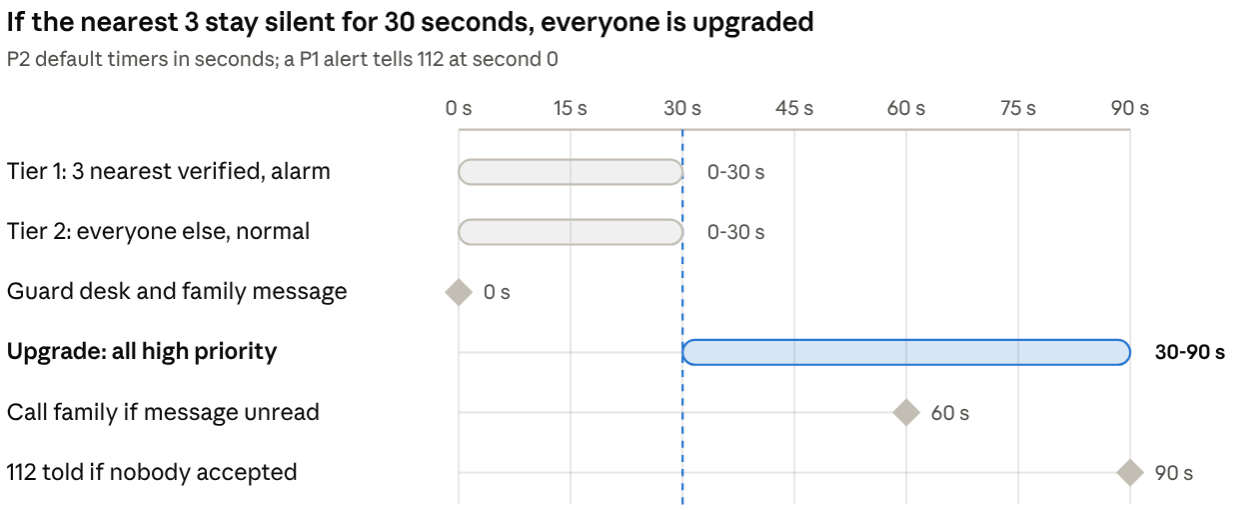
v1 locked the incident to the first volunteer. v2 allows a small team per case type, then tells the rest "covered, stand by":

Case	Responders accepted	Why
Medical, collapse, possible cardiac arrest	Up to 3 (one calls 112, one does CPR, one fetches an AED or guides the ambulance)	Hands-only CPR is exhausting; two people can swap
Fall or general medical	1 primary plus 1 backup	Backup steps in if the primary is delayed
Child or cognitive disability	1 verified primary plus the guard	Avoid crowding a frightened person
Harassment or violence	Up to 3, "observe and call" only	Safety in numbers; never confront

All numbers are proposals. **Decision needed.**

9. Tiered geofenced alerting

Everyone nearby is alerted at the same moment, but not equally. The 3 nearest verified volunteers get a high-priority "you are closest" alarm; everyone else gets a normal alert; if none of the 3 responds within 30 s, everyone is upgraded to high priority.



first 90 seconds of an alert · 6 events

Both tiers, the guard and the family are alerted at second 0. At 30 s everyone is upgraded, at 60 s an unread family gets a call, and at 90 s 112 is told (Section 8.4).

9.1 The tiers

Tier	Who	Alert style	Message
Tier 1, high priority	The 3 nearest **verified** volunteers who are available	Full-screen alarm, siren sound, repeats until answered	"You are in close proximity. Help the person. Medical, senior, about 300 m."
Tier 2, normal	Every other available volunteer in the radius, verified or not	Standard notification	"Help needed nearby, about 900 m. Tap to view."
Upgrade at 30 s	If no Tier 1 volunteer has accepted	All Tier 2 alerts re-sent as high priority	"Urgent: nearest helpers have not responded."

9.2 Rules that make it work

- **Rolling top 3.** When a Tier 1 volunteer taps Decline, the next nearest verified volunteer moves into Tier 1 straight away. GoodSAM does something similar: an unanswered alert disappears after 15 s and the next responder is tried (NIAS FAQ).
- **Early upgrade.** If all 3 decline before 30 s, upgrade everyone at once rather than waiting.

- **Responding means Accept or Decline.** Silence counts as no response. Decline asks for a one-tap reason (too far, busy, not safe, not trained) so dispatch learns.
- **Enough is enough.** Once the case has its responders (Section 8.5), everyone else gets "Covered, thank you, stay available".
- **Sparse areas.** If fewer than 3 verified volunteers are inside the radius, widen it in steps (for example 1.5 km, then 3 km, then 5 km; est.) and lean harder on the guard and family.
- **Protected profiles.** For a child, a person with a cognitive disability or a discreet case, an unverified volunteer can join only alongside a verified volunteer or the guard, never as the only responder. **Decision needed.**
- **Fatigue and fairness.** Cap alerts per volunteer per day, respect quiet hours unless the volunteer is on duty, and rotate among volunteers who are equally near.

When the person moves (Go pendant). The search circle is centred on the latest location and widened by its accuracy radius. If the person moves more than about 200 m (est.), candidates are re-ranked, the new nearest volunteers are alerted, and responders see the new spot.

9.3 Ranking: who is "nearest"

Distance alone picks the wrong person when a nurse is 50 m further away than a student. The ranking uses an effective ETA:

```
\text{score} = \text{ETA}_{\text{seconds}} - \text{skill bonus} -
\text{reliability bonus} + \text{language or preference penalty}
```

- **ETA:** from the volunteer's latest known position (Section 11) to the person's latest location (Section 18); walking speed under 1 km, two-wheeler above, if the volunteer said they ride one.
- **Skill bonus:** for example 60 s for a doctor or nurse on a medical case, 30 s for a certified first-aider (est.).
- **Reliability bonus:** small, from past accept and arrive rates.
- **Preference penalty:** if the profile asks for a woman responder first, or the volunteer does not share the person's language.
- **Verified first:** Tier 1 is taken only from verified volunteers. Unverified volunteers are ranked within Tier 2.

The weights are guesses to be tuned with pilot data. Every dispatch decision is logged so it can be explained later.

10. Face B: the volunteer app

Volunteers sign up on purpose, own smartphones and can be trained once, so they get a native app built around one job: hear the alert, decide in seconds, get there safely, and help well. The "no app" rule applies only to the person in trouble.

10.1 Screens and what each does

Screen	What it does
Registration	Identity, government ID, occupation and skills, certificates, address, consent (Section 12)
Setup checklist	Walks through every permission the alerts need (10.3) and ends with a test alert
Tutorial	Mandatory app walkthrough and simulation before the first real alert (Section 13)
Home	Big **I am available** toggle with modes (10.2), location freshness, trust badge, alerts answered
Alert	Tier 1: full-screen alarm. Tier 2: normal notification. Shows category icon, rough distance and ETA, age band, how many others are responding. Slide to accept (prevents pocket accepts) or Decline with a reason
Safety check	For harassment or violence: "Do not confront. Observe from a safe distance and call 112. Accept?" For a stealth alert, also: "Arrive quietly; do not mention SafeCircle or the alert out loud."
Mission	After accepting: map and directions to the person's live position with an accuracy circle (Section 18), full address, flat, landmark, gate notes, photo, handling banner, smart flashcards (Section 14). Roadmap: a Bluetooth warmer-colder finder and a make-it-beep button (off for discreet profiles)
Status bar	On my way, Arrived (suggested automatically near the spot), Need backup, Handed to ambulance, Resolved with outcome, Can't attend

Screen	What it does
Contact	One tap to call family, guard or 112. Calls go through number masking so neither side sees the other's personal number
Responder chat	Short group thread with other responders and the guard for this incident only; deleted after closure
I'm in danger	A volunteer's own SOS: alerts the guard, other responders and 112 with the volunteer's live location
After the call	Mission log, a wellbeing check ("That was a hard call, want to talk to someone?"), thanks from the family
Learn	Tutorial library, offline flashcards, CPR metronome practice, refresher quizzes
Profile	Skills, certificate expiry, verification level, pending community votes (Section 12), data download and delete
Settings	Max distance willing to travel, walk or two-wheeler, case types opted out of, languages, quiet hours, battery mode

10.2 The "I am available" toggle

Mode	Meaning	Location behaviour (Section 11)
Off	No alerts	None
Available now	Alerts until switched off	Low-power zone tracking plus wake-up on alert
Available until a time	For example "until 10 pm"	Same, auto-off at the time
Duty windows	Repeating schedule, for example weekday evenings	Active only inside windows
Home only	Alert me only when I am at home	No tracking at all: the registered home address is used while the phone sees home Wi-Fi

Mode	Meaning	Location behaviour (Section 11)
Auto-pause	Turns off below a set battery level (for example 15%) or while driving	Stops all location use

The app shows a plain line under the toggle: "While available, SafeCircle checks your approximate area about once an hour and your exact location only when someone near you needs help."

10.3 Setup checklist (permissions)

- **Notifications** on, with the SafeCircle alarm channel set to sound.
- **Full-screen alerts (Android 14+).** Only calling and alarm apps get this by default; others must ask the user (Google Play policy). The app asks during setup.
- **Critical Alerts (iPhone).** Breaking through silent mode needs a special Apple entitlement approved only for urgent personal health and public safety alerts (Apple forums). Apply for it; fall back to time-sensitive notifications if refused. **Verify.**
- **Location "Allow all the time"**, explained honestly (Section 11).
- **Battery optimisation off for SafeCircle.** Many phone brands popular in India stop background apps aggressively; the app shows brand-specific steps.
- **Test alert** to prove it works on this phone before going available.

10.4 Guards use the same app

Guards get a "guard" role: shift on and off instead of availability, the full flat number and access notes immediately (trusted role), and gate tasks ("open gate for ambulance"). A guard desk can also run the web dashboard with a loud alarm (Section 15).

10.5 Build path

Stage	Build	Why
Hackathon demo	Web app (PWA) in the phone browser	One codebase, no store approval, location works while open
Pilot	Native app in Flutter or React Native	Reliable wake-up pushes, background location, alarm sounds, full-screen alerts

11. Volunteer location and battery: the wake-up design

Yes, a wake-up mechanism is possible on both Android and iPhone, so "I am available" does not need constant GPS. The app keeps only a rough, low-power idea of the volunteer's area (refreshed about hourly or when they move), and takes one exact reading only when an alert near them is raised. Live GPS runs only after they accept, until they arrive.

11.1 Five layers, from cheapest to most precise

Layer	When it runs	How	Battery cost	Precision
L0 Home shortcut	Phone is on the volunteer's home Wi-Fi	Use the registered home address; no GPS at all	Near zero	Address
L1 Movement trigger	Volunteer moves a significant distance	OS-managed, low-power movement events (iPhone significant-change monitoring; an Android geofence around the last zone)	Very low	Neighbourhood
L2 Hourly refresh	About every 60 min while available, skipped if the phone has not moved	Android: a periodic background job using Wi-Fi and cell positioning, not GPS. iPhone: relies on L1, because iOS does not run apps on a fixed timer	Low	About 100 m to 1 km (est.)
L3 Wake-up on alert	An alert is raised and this volunteer's zone is within the search radius	Server sends a silent wake-up push; the app takes one fresh fix and reports it	One fix per alert	GPS-level
L4 Live en route	After Accept, until Arrived	Foreground location every few seconds, then GPS stops on arrival	Minutes per incident	GPS-level

GoodSAM shows the same trade-off: its accuracy slider uses cell triangulation, described as under 1% of battery, while GPS is called power hungry and not recommended; GPS also stops once the responder reaches the scene (NIAS FAQ).

11.2 How the wake-up works on each platform

- **Android:** a high-priority Firebase Cloud Messaging message is one of the allowed exceptions for starting a foreground service from the background. A service of type `location`, started by an app holding background location permission, can read location even while the app is in the background (Android developers). So: high-priority push, start a short location service, take one fix, send it, stop. Needs "Allow all the time" location and the Play Console background-location declaration.
- **iPhone:** the Location Push Service Extension (iOS 15+) is described by Apple as a power-efficient way to query a device's location even when the app is not running, triggered by a special `location` push from our server (Apple docs). It needs a restricted Apple entitlement and "Always" location permission. Developers report it may not wake on some locked devices (Apple forums), so L1 remains the fallback. **Verify** entitlement approval for a volunteer-safety app early.
- **Web app (demo):** a PWA cannot read location in the background. The demo uses location only while the app is open, and says so.

11.3 Never wait for the fresh fix

1. At 0 s the server ranks candidates from their last known zone and sends alerts at once.
2. In the same moment it sends wake-up pushes to every candidate within the radius plus a margin.
3. Fresh fixes arrive over the next few seconds and re-rank Tier 1 (the rolling top 3).
4. A phone that does not answer keeps its old zone, shown with its age ("location 50 min old"), and drops in rank.

This is why hourly is enough: as you noted, people usually stay in one place for a while, and the alert-time fix corrects anyone who has moved.

11.4 Privacy rules for volunteer location

- The server stores only a coarse zone (about 1 km cell) between incidents; never a movement history.
- Exact fixes are kept only for an incident and deleted shortly after it closes (for example within 72 h; **Decision needed**).
- Family and the person see the volunteer's name, verified badge, distance and ETA, never a live map of the volunteer.
- The volunteer can see and clear their own location log.
- Spoofed GPS: detect mock-location settings, cross-check against network position; device integrity checks (Play Integrity, App Attest) are Roadmap. Not perfect.

12. Volunteer trust: ID, occupation, verified tag and community vote

A volunteer earns the **Verified** tag only after a government ID check plus approval by their own community, either a society verifier or a vote of nearby verified members. Unverified volunteers can still help, but only in Tier 2 and never alone with a child or a person with a cognitive disability.

12.1 Trust levels

Level	Badge	How it is earned	What it unlocks
L0 Registered	None	Phone one-time code, selfie, profile, code of conduct, mandatory tutorial	Tier 2 alerts; protected-profile limits (Section 9)
L1 ID-checked	"ID checked"	Government ID shared with consent through DigiLocker (Aadhaar, PAN or driving licence), selfie matched to the ID photo	Same as L0, plus eligible for a community vote
L2 Verified Member	**Verified**	L1 plus society-verifier approval or a passed community vote (12.3)	Tier 1 high-priority alerts; can vote on new members
L3 Professional	Skill badge (Doctor, Nurse, Paramedic, First-aider)	Credential checked (12.4)	Ranking bonus for matching cases (Section 9.3)
L4 Police-verified	"Police verified"	Formal police verification through a partnership	Roadmap: eligible for 112-linked programmes

12.2 What registration collects

- Name, photo, phone, date of birth (18+ only), languages, home and optional work address.
- **Occupation** (doctor, nurse, paramedic, pharmacist, police, fire, ex-serviceman, security, teacher, student, other) and any first-aid or CPR certificate with its expiry date.

- **Government ID** via DigiLocker. KYC providers offer a consent flow where the user logs in to DigiLocker and approves sharing, and documents such as Aadhaar, PAN and driving licence are fetched only after that consent (Cashfree docs).
- **What we keep:** the verification result, name match, provider reference and a masked ID (last 4 digits). Not the full Aadhaar number or a copy of the card. **Verify** Aadhaar storage rules with someone qualified.
- Transport (walk, cycle, two-wheeler), maximum distance, case types opted out of, emergency contact for the volunteer.

12.3 Community vote ("vouching")

The question is "do you know and trust this person?", not "do you like them?".

1. A new L1 volunteer requests verification for their area.
2. The system invites the society verifier (if the society has one) and up to about 15 randomly chosen L2 members living within about 1 km (est.).
3. Each voter sees only: photo, first name and initial, street or block (not flat), occupation, and how long they have lived there. Never the ID.
4. Voters choose: **I know them and vouch**, **I know them and have a concern** (reason goes privately to the admin), or **I don't know them** (abstain).
5. **Pass rule (proposal):** at least 3 vouches, and vouches are at least two thirds of non-abstain votes, within 7 days. Any concern sends the case to an admin instead of auto-rejecting. **Decision needed.**
6. **Society verifier path:** an RWA office-bearer appointed by the society admin can approve residents and staff they know directly. This is the fallback when there are not enough voters nearby.
7. **Quorum not reached:** the volunteer stays L1 and is told how to reach a verifier. They are never shown as rejected.

Safeguards

- **Collusion:** accounts sharing a phone, device or address cannot vouch for each other; each voter has a monthly vouch limit; new L2 members cannot vote for 30 days.
- **Accountability:** if a vouched volunteer is suspended for misconduct within 6 months, the vouchers' future votes count for less.
- **Bias:** neighbourhood votes can reflect prejudice (for example against tenants, domestic workers, or by caste, religion or gender). Concerns need a written reason, every rejection can be appealed, the verifier path stays open, and rejection rates are monitored by group.

12.4 Checking professional claims

Claim	Check
Doctor	Registration number searched in the NMC Indian Medical Register by name, number, year and State Medical Council (guide)
Nurse, paramedic	State council registration or employer ID, checked by an admin
First-aid or CPR certificate	Upload, admin check, expiry date tracked; badge removed when it lapses, as GoodSAM does
Police, fire, ex-servicemen	Service ID, admin check

Occupation also routes cases: a doctor nearby moves up for a medical alert (Section 9.3). It never obliges anyone to respond.

12.5 Misconduct and accountability

- Every volunteer accepts a code of conduct: stay within your training, never take a child anywhere, never enter alone where a guard is available, hand over to the ambulance, no photos or posts.
- Every view of a person's profile is logged (who, when, which incident).
- After each incident the family and guard can rate and report. Reports lead to review, suspension or removal.
- The verified ID on record allows a police complaint if a volunteer causes harm.
- Re-verification every year; certificate badges expire automatically.

13. Volunteer training and tutorials

No volunteer can switch to "available" until they finish a short mandatory course and a practice alert. A library of emergency tutorials then stays in the app, offline, in Tamil, Hindi and English.

13.1 Mandatory onboarding course (about 30 minutes, est.)

Module	What it teaches	Check
M1 How SafeCircle works	Tiers, the 30-second upgrade, statuses, what the person and family see	3 questions

Module	What it teaches	Check
M2 Using the app	Accept, Decline, navigate, On my way, Arrived, Need backup, masked calls	Tap-through demo
M3 Rules and privacy	No photos or posts, data stays in the app, child rules, code of conduct	3 questions
M4 Your own safety	Check for danger first; never confront; when to wait for police; the "I'm in danger" button	3 questions
M5 Talking to the person	Introduce yourself ("My name is..., I'm a SafeCircle volunteer, help is on the way"); speaking with seniors, deaf people, autistic people and people with dementia	Scenario quiz
M6 Practice alert	A full simulated alert from siren to Resolved, including a flashcard	Must complete

Pass mark about 80% (est.). GoodSAM uses the same pattern: read the code of conduct and run a simulation before the first alert (St John poster).

13.2 Emergency tutorial library

Each topic is a 2-minute video, an illustrated card (the same card used as a flashcard in Section 14) and a short quiz.

Group	Topics
Life-threatening	Hands-only CPR with an in-app metronome; using an AED; choking (adult, child, infant); severe bleeding; unconscious but breathing (recovery position); severe allergic reaction
Medical	Stroke signs (BE FAST); suspected heart attack; seizure; low blood sugar; asthma attack (help with their own inhaler); fainting
Injury	Fall in an older person (do not lift, suspect a hip fracture); head injury; burns; electric shock (switch off first); fracture
Local risks	Heat stroke; drowning; snakebite (keep still, immobilise, get to hospital; no cutting, sucking or tourniquet); dog bite

Group	Topics
People and behaviour	Lost child (stay in a visible public place, call the parent, never take the child home); wandering person with dementia; autistic meltdown (reduce noise and touch); panic attack; harassment as a bystander (distract, delegate, document, delay, direct); domestic violence (do not intervene physically; call 112)
Building	Fire (evacuate, no lifts); gas leak (no switches, open windows); stuck lift
After a hard call	Normal stress reactions, when to talk to someone

CPR numbers used: compressions at 100 to 120 per minute, at least 5 cm (2 in) and no more than 6 cm (2.4 in) deep for an average adult; untrained rescuers do compression-only CPR (American Heart Association). The same AHA guidance notes that mobile apps alerting rescuers to a nearby arrest may raise bystander CPR rates. **Verify** against the newest AHA, ILCOR or Indian Resuscitation Council guidance before publishing.

13.3 Quality and honesty rules

- All medical content comes from a recognised body (for example Indian Red Cross, St John Ambulance, AHA, ILCOR) and is signed off by a doctor, with a review date on each card.
- The app course does **not** replace hands-on CPR training. Partner with the Red Cross, hospitals or medical colleges for monthly practical sessions; a practical certificate earns the First-aider badge (Section 12).
- Refresher micro-quizzes monthly; full refresh every 12 months.
- Learning can earn badges, but response speed is never gamified, so nobody rushes or takes risks to win points.

14. Smart flashcards at acceptance

The moment a volunteer accepts, the app predicts the 3 situations they are most likely to find, from the person's profile, age and own incident history. The volunteer taps the one that matches what they see and gets that card's steps at once. A "Not breathing? Start CPR" button is always on screen, whatever the prediction says.

14.1 What the volunteer sees, in order

1. **Safety line:** "Check for danger before you go in."

2. **Handling banner** from the profile, for example "Hard of hearing: face her, speak slowly" or "Autistic: do not touch, keep voices low".
3. **Likely situations:** three large chips, for example **Fall**, **Low blood sugar**, **Stroke**.
4. **Always visible:** "Not breathing or unresponsive? Start CPR" (opens the CPR card with metronome).
5. **Something else:** a symptom picker (not breathing, unconscious, bleeding, chest pain, can't speak, confused, fell, scared or upset, other), which opens the right card.

Tapping a chip opens its card: what to check, numbered steps, what not to do, when to call 112, and what to tell the ambulance crew. Cards are large-text, can be read aloud hands-free, and are cached on the phone so they work with no signal.

14.2 How the 3 situations are ranked

Signal	Example	Weight (est.)
Profile conditions	Diabetic, so low blood sugar; epilepsy, so seizure; heart disease, so chest pain or arrest; asthma; dementia, so confusion or wandering	High
This person's own history	4 of her last 6 incidents were falls	High
Trigger context	Check-in call answer ("2 fall"); caregiver's words; time of day	High when present
Age band	Over 75: fall, stroke, chest pain. Child: lost, injury, asthma	Medium
Everyone's history	What incidents for this age band usually turn out to be, learned over time	Low at first

```
\text{score}(c) = w_1 \cdot \text{profile}(c) + w_2 \cdot \text{history}(c) + w_3 \cdot \text{context}(c) + w_4 \cdot \text{age}(c) + w_5 \cdot \text{population}(c)
```

Start with hand-set weights and simple rules; replace them with a learned model only once pilot data exists.

Worked example. Lakshmi, 78, diabetic, hard of hearing, two falls in the last year, pressed at 6 am. Ranked: 1 Fall, 2 Low blood sugar, 3 Stroke. Banner: "Hard of hearing: face her, speak slowly. Key with neighbour in B-205."

14.3 The tap feeds back

- The chosen card becomes the incident's **volunteer-observed type**. The dashboard, the family and the 112 handoff see it, worded carefully: "Volunteer reports possible low blood sugar."
- The volunteer can change their choice as things develop; every change is logged.
- After closure the outcome is added to the person's history, which sharpens the next prediction.

14.4 Safety limits

- Cards are first-aid guidance, not a diagnosis. Each says so in one line.
- No medicine doses. Cards only cover helping a person use their **own** prescribed items (inhaler, glucose, adrenaline auto-injector) and only as the source guideline allows.
- Condition names are shown only to the accepted responder (Section 20).
- Every card is doctor-reviewed and dated (Section 13.3).

15. Family, guards and authorities

The family sees a live status page, the guard gets a loud desk alarm with the flat number, and 112 receives a ready-made handoff packet. Nobody has to phone around to find out what is happening.

15.1 Family and caregivers

- **Channel:** WhatsApp utility template with a link to a live status page (no app), plus SMS fallback. An automated call goes out if the message is not opened within about 60 s, then the next contact is tried.
- **The status page shows:** alert received; who accepted (name, Verified badge, skill such as "Nurse"); distance and ETA; arrived; handed to ambulance; resolved. Never the volunteer's live position on a map. The family does see the person's own location from the pendant.
- **Family actions:** "I'm on my way" (joins the responder list), call the volunteer or guard through a masked number, "This is a false alarm" (runs the verification in Section 6, it does not cancel on its own), and share the medical summary with the hospital.
- **Cost:** volunteer and family alerts are business-started WhatsApp messages, so they are paid utility templates (Section 4, T2).

15.2 Society security guards

Guards are often the fastest responders because they are on site and know the building.

- **Guard desk:** web dashboard with a loud alarm showing incident type, block and flat.
- **Guard phone:** the guard role of the volunteer app (Section 10.4), with SMS or voice-call fallback for basic phones.
- **Access:** guards see the full address and access notes immediately (trusted role).
- **Tasks:** go to the flat, open the gate and hold the lift for the ambulance, meet volunteers at the gate, do the in-person check after a cancel (Section 6).
- **Unreachable guard:** alert moves to the next guard, then the society admin; timers keep running.

15.3 Authorities: dashboard and 112 handoff

- **Dashboard** (society desk or police control room): live map with the person's position, its source and accuracy, incident list, volunteer and guard status, timeline of every event, and the escalation state. Stores the incident log for review.
- **112 handoff packet:** person's name and age band, exact address and live coordinates with their source and accuracy, profile summary relevant to care, volunteer-observed type (Section 14), who is on scene, and callback numbers.
- **Demo:** the handoff is simulated on the dashboard and labelled as such.
- **Roadmap:** ERSS-112 already lists IoT-based and external signals among its channels (Section 3.1), so a formal integration with the state PSAP is a realistic pilot goal, subject to permission.
- **RWA report:** a monthly summary per society (alerts, response times, false alarms, volunteer coverage gaps).

15.4 Example timeline

Time	What happens
0:00	Lakshmi (78) holds her pendant. It buzzes once: sending
0:03	Server confirms. Pendant: blue pulse, two-tone chime. Guard desk alarm, family WhatsApp, 3 nearest verified volunteers get high-priority alarms, 9 others normal alerts
0:05	Wake-up pushes return fresh locations; a nurse 280 m away moves into the top 3
0:18	The nurse accepts. Pendant: green pulse, "help is coming" chime. Family page: "Priya (Verified, Nurse) is on the way, about 3 min"

Time	What happens
0:20	Other volunteers: "Covered, thank you" (one backup kept for a medical case)
0:25	Priya's flashcards: Fall, Low blood sugar, Stroke; banner "Hard of hearing"
3:10	Priya arrives (pendant: green solid). Guard has opened the door with the spare key from B-205. She taps **Low blood sugar**
4:00	Lakshmi is conscious and given sugar; family arrives; Resolved. If nobody had accepted by 1:30, 112 would have been told

16. Scenario playbook

These are the situations a real deployment will meet, including the ugly ones, and how the design handles each. Use them to test the build and to answer judges.

16.1 Medical and everyday emergencies

#	Scenario	Design response
1	Senior falls at home, conscious, cannot get up	Standard P2 flow; flashcard "Fall: do not lift"; guard brings the spare key
2	Senior presses, then collapses; nobody answers the check-in call	Raised to P1: everyone high priority, 112 told at once
3	Cardiac arrest in the society garden; a neighbour sends "HELP" or scans a pendant QR	P1; up to 3 responders (CPR, swap, fetch AED); metronome card
4	Diabetic senior confused at dawn	Flashcards rank low blood sugar first from the profile
5	Person found fine; refuses help	Volunteer marks Resolved with "no help needed"; family told; logged

16.2 Children and people with cognitive disabilities

#	Scenario	Design response
6	Child lost near school presses the bag-tag button	Pendant GPS gives the spot; verified responders only as primary; "stay in a visible public place, never take the child anywhere"; parent gets the responder's name and badge
7	Child presses repeatedly for fun	Each cancel is verified with the parent; history shows the pattern; parent coached; alerts never silently ignored
8	Person with dementia walks out; a stranger scans the pendant QR	Pendant GPS shows where they are, with the bystander flow as backup; wandering card; family and nearest volunteers alerted
9	Autistic teen in meltdown in a mall; caregiver sends "HELP"	Caregiver trigger; calm handling banner ("no touch, low voices"); one responder plus mall security
10	Deaf person needs help	No voice anywhere: WhatsApp, vibration confirmation, "write or gesture" banner

16.3 Violence, harassment and coercion

#	Scenario	Design response
11	Woman harassed on the street, phone in bag	Stealth mode: a normal press with lights only and no sound (or the discreet profile's triple press, with no light at all); no call to her, live GPS to responders and 112; responders "observe and call", woman responder preferred; P1 so 112 is told at once
12	Attacker forces the person to cancel	Verification call; no answer or duress PIN keeps the alert alive at high priority; button stays quiet so the attacker is not alerted
13	Break-in at night	Guard and 112; volunteers told to stay out and observe only

#	Scenario	Design response
14	Domestic violence; the abuser is a registered family contact	"Never inform" list in the profile hides status from that contact

16.4 Misuse and abuse of the system

#	Scenario	Design response
15	Fake "HELP" from an unknown number to lure a volunteer	Unverified incidents go to the guard desk only, never to volunteers automatically
16	Stalker joins as a volunteer to learn addresses	Address only after accepting; protected profiles need verified responders; every profile view logged; accepting without moving is flagged
17	Volunteer accepts and does not move	"Still coming?" at 3 min, then re-opened to others; reliability score drops
18	Too many volunteers rush in	Team caps per case (Section 8.5); "Covered, thank you" to the rest
19	Spoofed GPS to look closer	Mock-location detection and network cross-checks (Section 11.4)

16.5 System and environment failures

#	Scenario	Design response
20	Wi-Fi or power is down when the button is pressed	Retries with the red pattern; a Go pendant falls back to 4G; Roadmap: LoRa and a local siren
21	Button battery slowly dies	Heartbeat flags low battery to the caregiver days ahead
22	3 am, nobody available	30 s upgrade, wider radius, guard and family calls, 112 at 90 s
23	Flood or cyclone: dozens of alerts at once	Mass-incident mode: dashboard triage, cluster by building, volunteers asked to check the registered vulnerable people near them

#	Scenario	Design response
24	Fire in the building	112 at once; guards use the society's list of residents who need help evacuating
25	Person presses while away from home (another city)	Go pendant sends GPS, Wi-Fi scan and cell location over 4G; a Home button can only flag "away from home" and ask the caregiver
26	Family lives abroad	Local contacts and volunteers carry the response; family follows the status page

16.6 Volunteer welfare

#	Scenario	Design response
27	Scene turns dangerous	"I'm in danger" button: guard, other responders and 112 get the volunteer's live location
28	Volunteer witnesses a death or serious injury	Wellbeing check after the call, debrief contact, option to pause alerts
29	Volunteer worried about legal trouble	Code of conduct, stay within training, mission log; legal position in Section 20

16.7 Location scenarios (v2.1)

#	Scenario	Design response
30	Inside a mall, metro or basement where GPS cannot lock	Wi-Fi scan estimate, known places, cell as last resort; accuracy circle shown honestly; guard or mall security asked to search
31	Child carried off in a vehicle	Pendant moving at vehicle speed on a child profile raises P1 (Decision needed); live track to 112 and family; volunteers told not to chase
32	Pendant battery flat outdoors	Low-battery pattern and caregiver message at 20%; the bystander QR works with no power

#	Scenario	Design response
33	No mobile signal (rural road, lift)	Store and retry; a saved Wi-Fi if near; the first message goes the moment signal returns
34	GPS puts the person across the road or on the wrong floor	Responders see source and accuracy; card prompts "call out and look around"; make-it-beep button (Roadmap)
35	A family member wants to use the pendant to track an adult	Location shared only during alerts and drills by default; any other mode needs the adult's own consent; access logged

17. Design by user category

The same system adapts to four groups through the profile: which trigger they carry, how the button talks back, and what the responder is told.

17.1 The four groups

Group	Main problem	Trigger	Feedback profile	Responder is told	Special rules
Children	May have no phone, may panic, cannot give an address	Go pendant (GPS, 4G) on wristband, bag tag or ID holder	Child: friendly colours and chime	Name, age, photo, parent's number, "this is a minor"	Verified primary only; never take the child anywhere; parent gets responder's ID; parent verifies every cancel
Seniors	Falls, medical events, living alone, poor eyesight and hearing	Home buttons in bathroom and bedroom; Go pendant when going out	Loud: strong buzzer and vibration	Medical notes, doctor, where the key is	Nurses and doctors ranked up; check-in call at the start
Physical disabilities	Cannot reach a phone or a small button; cannot hear or see feedback	Large palm or elbow button at bed and wheelchair height	Visual-first or touch-and-sound by sub-type	Mobility aids, how to move them safely	No flow requires speaking or hearing
Cognitive disabilities	May not understand, may wander, sensory overload	Single pictorial Go pendant; caregiver trigger	Calm: soft and predictable	Handling instructions written by the caregiver	A cancel from the person alone never closes an alert

Stealth mode for girls and women. Any girl or woman who goes out alone can use the Stealth profile (Section 7.2), including adults outside these four groups. The button confirms each stage with light only, so nobody nearby can tell help is on the way.

17.2 Physical disability sub-types

Sub-type	Problem	Feature
Mobility	Cannot reach a phone or press a small button	Large-surface button; voice trigger and sip-and-puff switch are Roadmap
Visual	Cannot find or read a screen	Tactile button, distinct sound and vibration counts, screen-reader-friendly portal
Hearing	Cannot hear a callback or siren	Light and vibration confirmation; WhatsApp instead of calls

Sub-type	Problem	Feature
Speech	Cannot explain on a call	One press sends everything; responder told to expect a non-verbal person

112 defaults to a voice call, which excludes deaf and speech-impaired users. No SafeCircle flow requires speaking or hearing.

17.3 Feature matrix

Feature	Child	Senior	Physical	Cognitive
Physical button	Yes	Yes	Yes (large)	Yes (pictorial)
Bystander QR	Yes	Yes	Some	Yes (wandering)
Caregiver trigger	Yes	Yes	Some	Yes
Check-in call at start	Parent instead	Yes	If they can use a phone	Caregiver instead
Cancel verification	Via parent	Call to self	Call or WhatsApp	Caregiver plus guard check
Smart flashcards	Yes	Yes	Yes	Yes, handling first
Fall detection	No	Roadmap	Some	No
Daily check-in	No	Roadmap	Roadmap	Roadmap
Priority responder	Verified adult	Doctor or nurse	Trained helper	Caregiver-briefed
Go pendant with GPS	Yes	When going out	If they go out	Yes (wandering)

18. Hardware and location: the Home button and the Go pendant

To find a person away from home, the button needs GPS, and GPS alone is not enough: the location also needs a way out, because home Wi-Fi is not there outdoors. So the outdoor device is an ESP32 with Wi-Fi, a GPS receiver and a 4G modem, and SafeCircle becomes two devices: a cheap Wi-Fi **Home button** that never moves, and a **Go pendant** that works anywhere.

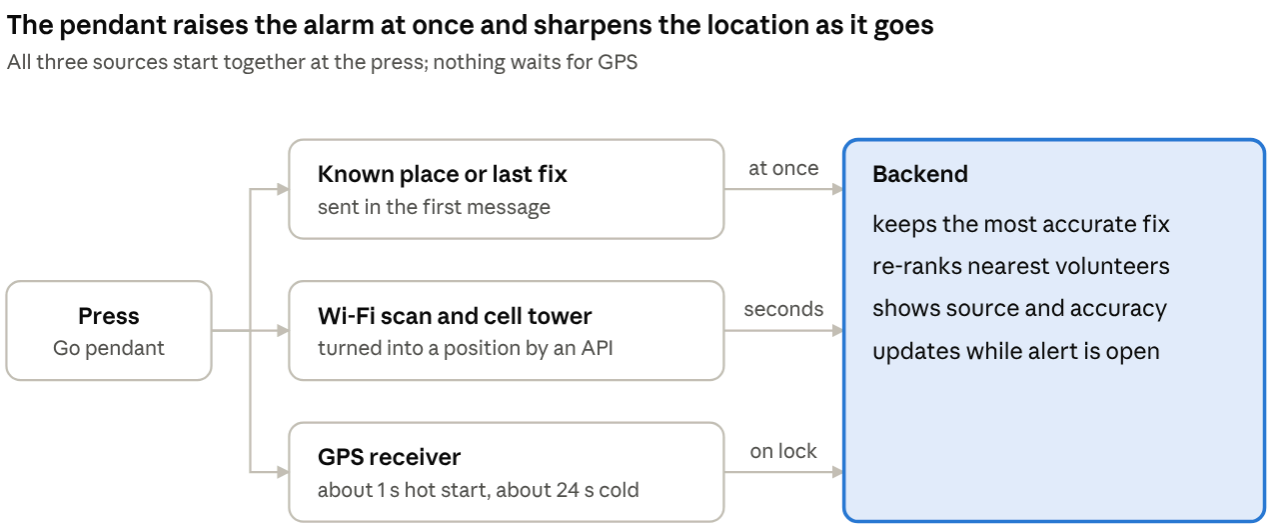
18.1 Two devices

	Home button	Go pendant
Where	Wall, bedside, bathroom, wheelchair	Worn outside: pendant, wristband, bag tag, school-ID holder
Inside	ESP32, button, RGB LED, passive buzzer, vibration motor	The same, plus a GPS receiver, a 4G modem with SIM, LTE and GPS antennas, a bigger battery
Location	Registered flat and room	GPS outdoors, Wi-Fi scan indoors, cell tower as backup, known places, registered address at home
Sends over	Home Wi-Fi	A saved Wi-Fi network if in range, otherwise 4G
Charging (est.)	Rarely: deep sleep between events	Every few days
For	Seniors at home, people with limited mobility	Children, seniors who go out, people who may wander

The hackathon demo can still use a Wi-Fi-only Home button. The demo Go pendant is an ESP32 plus a GPS module sending over the phone hotspot (18.6).

18.2 How the pendant finds the person: send now, refine later

The pendant never waits for GPS before raising the alarm. It sends the best location it has within seconds, then sharpens it.



pendant location sources · 3 in parallel, 1 backend

A later fix replaces the earlier estimate only when it is more accurate, so responders never see the location get worse.

1. **Press (0 s):** the ESP32 wakes and starts three things at once: the GPS receiver, the 4G modem, and a Wi-Fi scan. Listing nearby networks does not require joining any of them.
2. **First message (a few seconds, est.):** device ID, event, and whatever location it already has: the known place if it sees a saved network, the Wi-Fi scan list, the serving cell tower, and its last GPS fix with that fix's age. The backend turns the Wi-Fi and cell lists into a position and dispatches immediately.
3. **GPS fix:** about 1 s on a hot start, about 24 s on a cold start (18.3). Sent as a location update; dispatch re-ranks volunteers.
4. **While the alert is open:** an update about every 10 s while the person moves and about every 60 s while still (est.).
5. **Close:** GPS and modem switch off and the pendant goes back to deep sleep.

Source	Typical accuracy	Works where	Used when
Registered address	The flat	At home	Home button, or pendant on home Wi-Fi
Known place (saved Wi-Fi)	The building	School, day centre, relatives' homes	The pendant sees a saved network
GPS fix	About 1.5 m listed for a u-blox M10 chip-antenna module in open sky (SparkFun); worse near buildings	Outdoors, near windows	Main outdoor source
Wi-Fi scan	About 20 m in Google's own example	Indoors and in cities	GPS weak or not yet locked
Cell tower	About 180 m to 1.5 km in Google's examples (Google)	Anywhere with signal	Last resort
Bystander QR	The stranger's phone	Anywhere	No fix, or a flat battery

The backend always uses the most accurate fresh source and shows responders both the source and an accuracy circle ("GPS, about 8 m" or "Wi-Fi estimate, about 60 m"). It never hides uncertainty.

18.3 How GPS works, in plain words

- **Satellites and timing.** A GPS receiver listens to timing signals from satellites (GPS, plus Galileo, BeiDou, GLONASS or India's NavIC on multi-system receivers) and computes its position from them. It needs to see the sky; roofs, walls, basements and lifts block it.
- **Cold and hot start.** To compute a position it needs the satellites' orbit data. Without fresh data a cold start takes about 24 s on the u-blox MAX-M10S; with saved data a hot start takes about 1 s (SparkFun MAX-M10S).
- **Backup battery.** A small onboard battery keeps the clock and orbit data for up to about five hours with main power off (same source). A pendant that had a fix recently starts almost instantly.
- **Assisted GPS.** The pendant can download orbit data over Wi-Fi or 4G and skip the cold start; u-blox lists an aided start of about 1 s for its M10 chip-antenna module (SparkFun SAM-M10Q). Firmware refreshes this data while charging.
- **Power.** The MAX-M10S draws under 25 mW in continuous tracking, about 6-25 mA depending on mode (SparkFun MAX-M10S). The SAM-M10Q lists about 10 mA tracking and 5 mA in power-save mode (SparkFun SAM-M10Q). Small enough to run for a whole incident, too much to leave on all day.
- **Antenna.** The GPS antenna must face outward and up, away from the body and the battery. Test on a wrist, around the neck and in a pocket.
- **NavIC (optional).** India's own satellite system. Quectel's L89HA-S90 receives GPS and NavIC on L1 and L5 with an integrated antenna and assisted start, and is sold in India (Hubtronics). NavIC is required for vehicle trackers under AIS-140 (Telit); we found no such rule for personal devices.

18.4 How the alert leaves the pendant

1. **A saved Wi-Fi network** (home, school) if in range: fastest and free.
2. **4G Cat-1 through the SIM** everywhere else. Cat-1 runs on ordinary 4G networks. The LilyGO T-A7670G covers global LTE bands, including B1, B3, B5, B8, B40 and B41, and falls back to GSM (LilyGO). **Verify** band coverage with Jio, Airtel and Vi in the pilot area.
3. **Bluetooth to a paired phone** running a background helper (Roadmap).
4. **Store and retry** with the red pattern if nothing works.

- The LilyGO A7670 boards are data-only, with no voice calls (LilyGO). That is fine: the pendant never needs to call, and verification calls go to the person's phone or caregiver (Section 6).
- **SIM:** an IoT data SIM with a small plan; alerts and heartbeats use very little data. **Verify** plan prices.

- **Status back to the pendant** (Section 7.3) uses the same 4G link, over MQTT.

18.5 Power and battery

Mode	What is on	When
Sleep	Nothing but the ESP32's wake pin; the GPS backup battery keeps orbit data for hours	Almost all the time
Heartbeat	ESP32 and the cheapest link (saved Wi-Fi, else 4G) for a few seconds	About every 6 h (est.)
Alert	ESP32, GPS, 4G, Wi-Fi scan, LED, buzzer, vibration	Until the incident closes
Charging	Assisted-GPS data refresh, firmware updates	On the dock

- **GPS stays off between alerts.** That saves battery and makes the pendant a safety device, not a tracker (Section 20.6). An opt-in safe-zone mode is Roadmap.
- **Current peaks.** A 4G modem draws short, high current bursts while transmitting. The MT3608 boost converter from v1 is not suitable; use the board's own power path or a LiPo with a regulator rated for the modem's peak. **Verify** the peak in the modem datasheet. LilyGO's A7670 boards are designed around an 18650 cell.
- **Target (est.):** a charge every 3-7 days, a low-battery pattern on the pendant and a message to the caregiver at 20%.
- **Charging:** a magnetic or pogo-pin dock (Roadmap enclosure), so seniors and children never fiddle with a cable.

18.6 Module options

Route	What it is	Good	Catch	Price seen
A. Demo	ESP32 DevKit plus a GNSS breakout: u-blox MAX-M10S or SAM-M10Q, or a budget NEO-6M or ATGM336H	Cheapest, simple serial or I2C wiring	No 4G; outdoors it sends through the phone hotspot	SparkFun MAX-M10S about ₹6,040 imported (Tanotis); budget modules about ₹300-600 (est.)

Route	What it is	Good	Catch	Price seen
B. Pilot	LilyGO T-A7670G R2 with GPS: ESP32-WROVER-E, A7670G 4G Cat-1, LTE and GPS antennas, 18650 holder	Everything on one board, with LTE and GPS examples	Bulky (111 × 34 × 19 mm per LilyGO); data only	₹6,999 incl. GST (Probots)
C. Production	Custom PCB: small ESP32 (C3 or S3), Cat-1 modem, M10-class or NavIC GNSS, chip antennas	Small, much cheaper at volume	Design time, RF tuning, certification	**Verify** at quote
D. Budget variant	ESP32 plus GNSS, sending through Bluetooth to a paired phone	No SIM cost	Needs a phone nearby with the helper	Roadmap

Recommendation: route A for the hackathon, route B for the pilot, route C for scale. Check that any LilyGO board ordered is the GPS version: LilyGO sells it with and without GPS (LilyGO).

18.7 Firmware behaviour

- **Sleep and heartbeat** as in 18.5. The button pin wakes the chip.
- **Press timing:** about 2 s hold raises SOS; about 4 s requests cancel; triple press is silent SOS on discreet profiles (Section 6).
- **On press:** start GPS, modem and Wi-Fi scan together; send the first message without waiting for GPS (18.2).
- **During the incident:** send location updates (18.2), keep the status link open (Section 7.3), keep sending silently while a cancel is being verified (Section 6).
- **Retry:** queue messages, retry with back-off, red pattern if nothing gets through.
- **No personal data on the device.** Location is attached only to alerts, drills and heartbeats that report link health.

Example alert from a Go pendant (placeholder values):

```
{
  "device_id": "GO-021",
  "event": "panic",
  "battery": 76,
  "seq": 112,
  "location": {
    "gps": {"lat": 13.0827, "lng": 80.2707, "acc_m": 8, "age_s": 3},
    "wifi": [{"bssid": "a4:2b:..", "rssi": -61}, {"bssid": "f0:9f:..",
"rssi": -74}],
    "cell": {"mcc": 404, "mnc": 0, "tac": 0, "cell_id": 0},
    "known_place": null
  },
  "signature": "a91f3c..."
}
```

Events: `panic`, `cancel`, `silent_panic`, `location_update`, `heartbeat`, `low_battery`. The backend turns the Wi-Fi and cell lists into a position with a geolocation service such as Google's Geolocation API, which accepts the Wi-Fi access points and cell towers a device can see and returns a position with an accuracy radius (Google).

18.8 Bill of materials (prices est., confirm at purchase)

Home button (2 for the demo, unchanged from v2):

Item	Qty	Note	Approx. ₹
ESP32 DevKit (30-pin)	2-3	One main, one tested spare	700-1,400
30 mm arcade or panic push button	2	Momentary, normally open	100-200
Passive buzzer	3	Chimes and multiple tones	45-90
RGB LED, or a WS2812B LED or small ring	2-3	Ring allows the "fill while holding" effect	50-250
Coin vibration motor 3 V, transistor, flyback diode	2		100-150
LiPo 3.7 V 1000 mAh, TP4056 with protection, MT3608 boost, switch	2 sets	Fine for the Home button; not for 4G	600-900
Breadboards, jumpers, perfboard, wire, enclosure, strap, data cables, QR sticker	1 set		1,000-1,200

Go pendant prototype (1 for the demo):

Item	Qty	Note	Approx. ₹
ESP32 DevKit	1	Route A	350-700
GNSS module with antenna	1	Budget module, or MAX-M10S for low power	300-6,040
Same button, LED, buzzer, vibration parts as above	1 set		300-600
LiPo 1500-2000 mAh with a suitable regulator, or a power bank for the demo	1		400-800
Or, instead of the rows above: LilyGO T-A7670G R2 with GPS (route B) plus an 18650 cell and an IoT data SIM	1	Pilot route	6,999 plus cell and SIM

Tools if the team lacks them: soldering iron, solder, stripper, multimeter, glue gun (about ₹800-1,500).

18.9 Build order and test plan

1. Button and LED on a breadboard; a press prints to serial.
2. Long-press timing (2 s and 4 s) and triple-press detection.
3. Wi-Fi POST to a test endpoint, then to the real backend.
4. Passive buzzer tones, RGB patterns and vibration counts for every stage in Section 7.
5. Status polling after a press.
6. **GPS:** read fixes from the module over serial or I2C; log cold-start and hot-start times outdoors.
7. **Location in the alert:** add the GPS fix, Wi-Fi scan list and known-place match to the payload; the backend calls the geolocation service; the dashboard shows source and accuracy.
8. **Updates:** send location every few seconds while the incident is open; stop on close.
9. **4G (pilot):** attach the modem, send with Wi-Fi switched off, receive status over MQTT.
10. Battery and enclosure; rehearse the full flow at least five times.

Test plan: wrong signature rejected; no network shows the failed pattern and retries; double press makes one incident; cancel triggers verification; each stage plays the right pattern; **outdoor** cold and hot start times; **indoor** test where GPS fails and the Wi-Fi

estimate takes over; **walking** test where the dashboard pin follows; GPS fix compared with a phone's position; 24-hour battery drain in sleep; time from press to first volunteer alert recorded.

19. Technical architecture, data model and security

One backend owns every incident; every channel is just an input or an output around it. Privacy rules such as "address only after accepting" are enforced on the server, never only in the app.

19.1 Components

Component	Job	Suggested tech (Decision needed)
Ingest gateway	Button API (over Wi-Fi and 4G), MQTT broker, WhatsApp, SMS, missed-call and voice webhooks, bystander page	HTTPS endpoints; Mosquitto or EMQX for MQTT
Incident service	States, merging, severity, timers	Node (NestJS) or Python (FastAPI); job queue on Redis for timers
Dispatch service	Geo search, tiers, ranking, upgrades	PostgreSQL with PostGIS
Location service	Volunteer zones, wake-up pushes and fresh fixes; person location fusion from GPS, Wi-Fi scan, cell and known places; retention clean-up	Firebase Cloud Messaging (Android), APNs including location pushes (iPhone); a Wi-Fi and cell geolocation API
Notification service	App alerts, WhatsApp templates, SMS, voice calls, masked calls	FCM/APNs, WhatsApp Business provider, Exotel or similar
Verification service	DigiLocker checks via a KYC provider, credential review, vote engine	Provider API plus admin console
Content service	Tutorials and flashcards, versioned and dated	Static files plus a card database
Web apps	Setup portal, family status page, guard and authority dashboard	React or plain HTML; Leaflet with OpenStreetMap; WebSockets for live updates

Component	Job	Suggested tech (Decision needed)
Volunteer app	Face B	Flutter or React Native (PWA for the demo)
Audit log	Every view, dispatch and status change	Append-only table

19.2 Data model

- **people:** profile, address, pin, key location, medical and handling notes (encrypted), language, feedback profile, safety rules, consent
- **caregivers** and **sender_numbers:** phone, role (`self`, `caregiver`), verified flag, can-cancel flag, never-inform flag
- **devices:** device_id, person_id, secret key, claim code, status, last_seen, battery, Wi-Fi network fingerprint, type (home or go), SIM ID, known places, last GPS fix time
- **societies** and **guards:** society, guard phone, on-duty status, admin
- **volunteers:** trust level, skills, credentials with expiry, availability mode, duty windows, reliability score, transport, max distance
- **volunteer_zones:** volunteer_id, coarse cell, updated_at (no history)
- **volunteer_fixes:** volunteer_id, incident_id, lat, lng, accuracy, time (deleted after retention)
- **votes:** candidate, voter, choice, private reason, time
- **incidents** and **incident_events:** source, severity, state, flags (escalated, duress, unverified, drill), every event with a timestamp
- **dispatches:** incident, volunteer, tier, sent, response, reason, upgrade time
- **cards** and **card_selections:** card content and version; which card a responder chose and when
- **audit_log:** who viewed which profile, when, for which incident

person_locations (new in v2.1): incident_id, lat, lng, accuracy, source (gps, wifi, cell, known_place, registry), speed, time. Deleted on the same schedule as volunteer fixes (Section 20.2).

19.3 Key endpoints

Endpoint	Purpose
`POST /api/sos/device`	Button events (signed), with a location object from the Go pendant (Section 18.7)

Endpoint	Purpose
`GET /api/device/{id}/status`	Button polls its incident stage
`POST /webhooks/whatsapp`, `/webhooks/sms`, `/webhooks/voice`	Provider callbacks (signature checked)
`POST /api/volunteer/availability`	Toggle and mode
`POST /api/volunteer/location`	Zone update or incident fix
`POST /api/incidents/{id}/respond`	Accept or Decline with reason
`GET /api/incidents/{id}/briefing`	Address, profile, cards; returns 403 unless this volunteer accepted
`POST /api/incidents/{id}/status`	On my way, Arrived, Need backup, Resolved
`POST /api/votes`	Community vote
`WS /ws/dashboard`	Live incident stream
POST /api/device/{id}/location	Location updates from the Go pendant while an incident is open

19.4 Security

- **Device spoofing and replay:** HMAC signature with a per-device key, plus an increasing sequence number.
- **Transport:** TLS everywhere in production. Plain HTTP on the hotspot is demo-only and labelled so.
- **At rest:** field-level encryption for medical notes, access codes, ID references and person location fixes.
- **Staged disclosure** enforced by the briefing endpoint, not by hiding fields in the app.
- **Webhooks:** verify every provider signature; reject unsigned calls.
- **Rate limits** per device, phone number and volunteer.
- **Masked calling** so no phone numbers are exchanged.
- **Retention jobs** delete fixes, chats and recordings on schedule (Section 20).

20. Privacy, legal and safety

Two legal facts shape the design: India's data protection rules now require verifiable consent for children and for persons with disabilities who have a lawful guardian, and India's statutory Good Samaritan protection covers road accidents only. Neither is a

blocker, but both must be designed for. This section is not legal advice; get a qualified review before a pilot.

20.1 Data protection (DPDP Act 2023 and Rules 2025)

- The DPDP Rules were notified on 14 November 2025 with a phased rollout; the core duties (notice and consent, security, breach reporting, children's consent) apply 18 months later, around May 2027 (SCC Online, AMLEGALS).
- **Children:** Rule 10 needs verifiable consent from a parent before processing a child's data; Rule 11 needs a lawful guardian's verifiable consent for a person with a disability (SCC Online). The portal's consent step (Section 5.1) is built for this. Many adults with disabilities have no guardian and consent for themselves.
- **Design to the rules now** even though the main duties start in 2027: a pilot would run into that date.
- **Data minimisation:** every profile field must change a response (Section 5.2). Volunteers store a coarse zone, not a movement history (Section 11.4). The pendant's GPS is off between alerts (Section 18.5).
- **Rights:** download and delete for people, caregivers and volunteers; consent can be withdrawn in the portal or app.

20.2 Retention schedule (proposal, Decision needed)

Data	Kept for
Volunteer exact location fixes	72 h after the incident closes
Responder chat, masked-call logs	7 days
Incident record and outcome	1 year, then anonymised for statistics
Audit log of profile views	1 year
ID verification result (masked)	While the account exists
Person's profile	Until the caregiver or person deletes it
Person's location fixes from the Go pendant	72 h after the incident closes

20.3 Government ID

Use DigiLocker consent flows through a KYC provider and keep only the result, a masked number and the provider reference (Section 12.2). Do not store full Aadhaar numbers or card images. **Verify** with a lawyer which ID checks a non-regulated private platform may run.

20.4 Volunteer legal protection

- India's Good Samaritan guidelines came from the Supreme Court's 2016 *SaveLife Foundation* ruling and were later written into the Motor Vehicles Act as Section 134A in the 2019 amendment (Supreme Court order). That section protects people helping victims of accidents involving a motor vehicle (Daily Star explainer).
- **Gap:** most SafeCircle cases (a fall at home, a seizure, harassment) are not road accidents, so that specific protection does not apply. **Verify** with a lawyer what general-law protection exists for good-faith first aid.
- **Mitigations:** a code of conduct (stay within training, hand over to professionals, no driving under emergency conditions), the mission log as a record, volunteer accident insurance for the pilot, and clear terms that volunteering is optional. GoodSAM's approach is similar: no obligation to attend, and a code of conduct designed to keep responders within a Good Samaritan act (NIAS FAQ).

20.5 Platform honesty

- SafeCircle never promises that a volunteer will come. Every screen and the terms say it adds to 112, it does not replace it.
- Medical content is doctor-reviewed and dated (Section 13.3).
- Verification calls that are recorded say so at the start.

20.6 Ethics watch-list

- Community votes can carry prejudice (Section 12.3 safeguards).
- Wandering geofences and daily check-ins for children and people with dementia are surveillance; they stay Roadmap and opt-in with consent. The Go pendant is not a tracker: by default it shares location only during an alert or a drill, and any other mode needs the adult's own consent, or a parent's for a child.
- A discreet profile must never become a way to hide abuse of the person by the person who set it up; the "never inform" list is reviewed by an admin when it names a caregiver.

21. Feature catalogue and MVP

45 features, 19 of them Must. The Must set is the smallest build that shows the whole chain end to end and every new idea in this version. Change a tier with the dropdown.

Effort: S = a day or less, M = 2-4 days, L = a week or more (est., team of 3-4).

ID	Feature	Stage	Criteria	Effort	Tier
F1	Home button (Wi-Fi), 2 s hold	Trigger	C1 C5	M	Must

ID	Feature	Stage	Criteria	Effort	Tier
F2	Long-press cancel plus verification call (simulated in demo)	Trigger	C1	M	Must
F3	WhatsApp "HELP" with registry lookup	Trigger	C1 C5	M	Must
F4	Caregiver trigger	Trigger	C1 C5	S	Should
F5	Bystander QR on the pendant	Trigger	C1 C5	S	Should
F6	Silent SOS (discreet triple press)	Trigger	C1 C5	S	Should
F7	SMS keyword and missed-call trigger	Trigger	C1 C5	M	Roadmap
F8	Setup portal: pairing, profile, trusted numbers, consent	Setup	C2 C5	M	Must
F9	Browser Wi-Fi provisioning (SoftAP or Improv)	Setup	C5	M	Should
F10	Drill mode	Setup	C4	S	Should
F11	Unified incident pipeline with merging	Engine	C2	M	Must
F12	Severity rules P1-P3	Engine	C2	S	Should
F13	Timers and simulated 112 escalation	Engine	C2	S	Must
F14	Device signature and sequence check	Engine	C2	S	Should
F15	Check-in call at the start of an alert	Engine	C1 C4	M	Roadmap
F16	Tiered alerts: top 3 verified, 30 s upgrade	Alert	C3	M	Must
F17	Rolling top 3 and early upgrade	Alert	C3	S	Should
F18	Ranking with skill and reliability bonus	Alert	C3	S	Should
F19	Team caps per case type	Alert	C3	S	Should
F20	Volunteer app: alert, accept, status taps	Volunteer	C3 C4	M	Must
F21	"I am available" toggle (modes later)	Volunteer	C3	S	Must
F22	Wake-up location and battery layers (native app)	Volunteer	C3	L	Roadmap
F23	"I'm in danger" button	Volunteer	C3	S	Should
F24	Masked calling and responder chat	Volunteer	C3	M	Roadmap
F25	Trust levels and Verified tag	Trust	C3	S	Must
F26	Community vote and society verifier	Trust	C3	M	Should
F27	DigiLocker ID and credential checks	Trust	C3	M	Roadmap

ID	Feature	Stage	Criteria	Effort	Tier
F28	Mandatory tutorial and practice alert	Training	C3	M	Should
F29	Full emergency tutorial library	Training	C3	L	Roadmap
F30	Responder briefing and handling banner	Response	C5	S	Must
F31	Smart flashcards with tap feedback	Response	C3 C5	M	Must
F32	Staged disclosure enforced on the server	Response	All	S	Must
F33	Button status patterns (light, sound, vibration)	Feedback	C4 C5	M	Must
F34	Feedback profiles (loud, calm, discreet, stealth…)	Feedback	C4 C5	S	Should
F35	Family live status page	Feedback	C4	S	Must
F36	Guard and authority dashboard	Feedback	C2 C4	M	Must
F37	Heartbeat and low-battery alerts	Reliability	C2	S	Should
F38	Retry and failed pattern on the button	Reliability	C1	S	Must
F39	Go pendant: ESP32 plus GPS module (demo over the hotspot)	Trigger	C1 C5	M	Must
F40	Location fusion and live updates during an alert	Engine	C2 C3	M	Must
F41	Wi-Fi scan positioning through a geolocation API	Engine	C2 C5	S	Should
F42	Known places (saved, named Wi-Fi networks)	Setup	C2	S	Should
F43	4G Cat-1 link with SIM	Trigger	C1	M	Should
F44	Bluetooth finder and make-it-beep	Volunteer	C3	M	Roadmap
F45	Safe-zone mode (opt-in)	Engine	C5	M	Roadmap

Roadmap beyond the table: daily check-in for seniors living alone; fall detection; LoRa gateways with a society siren; real ERSS-112 integration; voice prompts and voice trigger in Tamil and Hindi; mass-incident mode; device attestation against fake GPS; gate integration.

21.1 Criteria coverage

Criterion	Delivered by
C1 Faster initiation	F1-F7, F15, F38, F39, F43
C2 Unified dispatch	F8, F11-F14, F36, F37, F40-F42

Criterion	Delivered by
C3 Volunteer coordination	F16-F29, F31, F40, F44
C4 Status updates	F10, F20, F33-F36
C5 Accessible without an app	F1, F3-F6, F8, F30, F31, F33, F34, F39, F41, F45

22. Demo plan, cost and time

The demo shows six short scenes in about 7 minutes, with timers compressed and labelled "demo speed". The prototype costs about ₹5,500-15,000 (est.), depending on the GPS module, and takes a team of 3-4 about 4-5 weeks part-time, or a hard-cut version in a 24-48 hour hackathon.

22.1 Demo scenes

Scene	What the audience sees	Proves
1. The press (2 min)	Senior's pendant: hold, blue "sending", chime "received". Dashboard lights up. Two volunteer phones get the full-screen "you are closest" alarm, a third gets a normal alert. Nobody answers, so at 10 s (demo speed for 30 s) the third phone upgrades. It accepts; pendant turns green with the "help is coming" chime; family page shows "Priya (Verified, Nurse), about 3 min"	Phone-free trigger, tiers, upgrade, feedback
2. The flashcards (1 min)	Volunteer phone shows the handling banner and three chips: Fall, Low blood sugar, Stroke. Tap Low blood sugar; the card opens and the dashboard shows "Volunteer reports possible low blood sugar"	Profile-aware response

Scene	What the audience sees	Proves
3. Forced cancel (1 min)	Long-press cancel. Dashboard shows the verification call; presenter clicks "no answer". Alert continues at high priority, flagged "possible duress", while the pendant stays quiet	Duress safety
4. No app at all (1 min)	Registered number sends "HELP" on WhatsApp: full incident appears. Unknown number: unverified flow to the guard only. A stranger scans the pendant QR and shares a location	Face A without hardware
5. Becoming verified (1 min)	New volunteer finishes the tutorial; two phones receive a vouch request and vouch; the badge turns Verified	Trust model
6. Outdoors (1 min)	Presenter takes the Go pendant prototype outside and presses it. The dashboard pin starts at the Wi-Fi estimate, then jumps to the GPS fix with a smaller accuracy circle; as the presenter walks, the pin and the volunteer's map follow	Locating the person anywhere

Say plainly on stage: the 112 handoff and verification call are simulated; the Go pendant sends over the phone hotspot instead of its own 4G; timers are compressed; the web app uses location only while open, and the production app uses the wake-up design in Section 11.

Demo kit: 2 Home buttons (one spare), 1 Go pendant prototype, 4 phones (2 volunteers, 1 family, 1 for WhatsApp and voting), laptop for the dashboard, phone hotspot plus a backup hotspot, printed QR, and a pre-recorded video fallback. Rehearse at least five times on the hotspot.

22.2 Prototype cost (est., verify at purchase)

Item	Cost (₹)
2 Home buttons plus 1 Go pendant prototype with a GPS module (Section 18.8)	3,500-10,000

Item	Cost (₹)
Tools if not owned	800-1,500
Hosting on free tiers	0-500
WhatsApp test number	0
Voice provider trial for the verification call (optional)	0-1,000
Maps (OpenStreetMap, Leaflet)	0
LLM API for optional triage	200-500
Printing (QR stickers, posters)	200
Total	**about 5,500-15,000**

22.3 Pilot cost (one society, 50-100 households, est.)

Item	Cost
About 60 Home buttons and 40 Go pendants (mix est.)	₹2-3 lakh
3-5 LoRa gateways (if used)	₹15-30k
Cloud, WhatsApp templates, SMS, voice calls and pendant SIM data	₹8-18k per month (est.)
ID checks through a KYC provider	per check; **Verify** pricing
App store accounts	Apple Developer Program USD 99 per year; Google Play USD 25 one-time
Volunteer accident insurance	**Verify** quotes
CPR practical workshops with a partner	Often free through NGOs or hospitals; **Verify**
Total	**about ₹3-4.5 lakh plus monthly costs**

A pilot takes 3-6 months, mostly field testing of range and battery, volunteer onboarding and building trust with the society and local police. Software is not the bottleneck.

22.4 Build time (team of 3-4, est.)

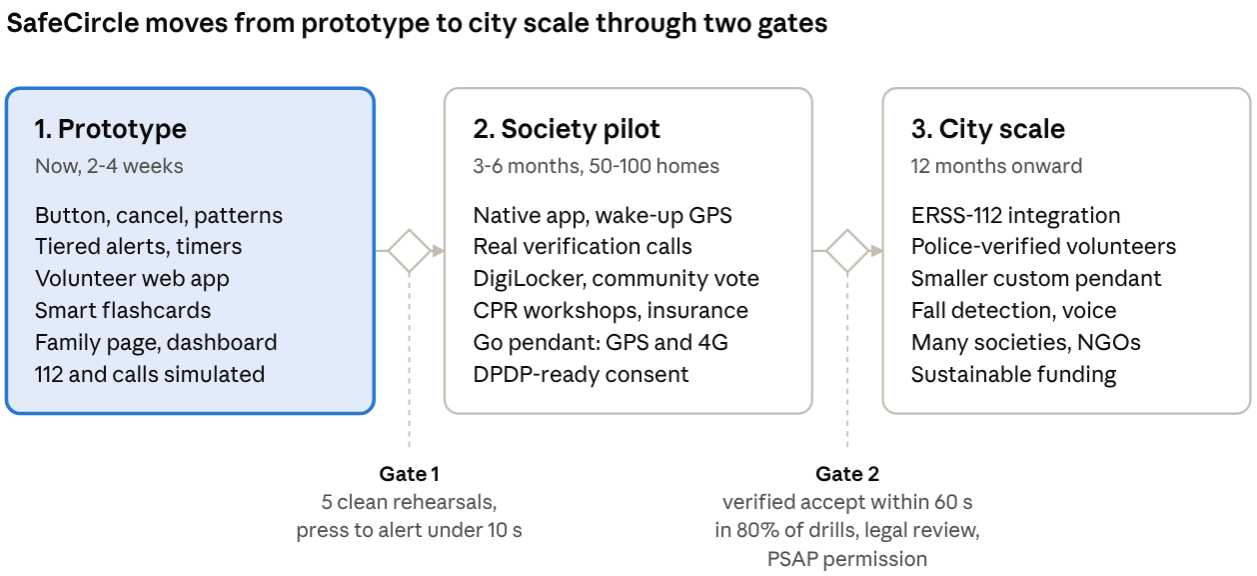
Component	Effort
Backend: pipeline, tiers, timers, WhatsApp webhook, location fusion	6-8 days
Setup portal and family status page	2-3 days
Volunteer web app: alerts, accept, status, flashcards, vote screen	5-6 days
Button and pendant firmware: timings, patterns, polling, GPS and location payload	6-8 days
Dashboard with simulated verification call and 112	3 days
Content: tutorial and 6 flashcards, doctor review	2 days
Integration, testing, slides	4-5 days
Total	**about 4-5 weeks part-time, or 2-3 weeks full-time**

24-48 hour hackathon cut: button with hold, cancel and four patterns (sending, received, accepted, failed); a GPS module on one button sending its fix with the alert; backend with tiers and demo-speed timers; volunteer web app with accept and three flashcards; family page; dashboard with simulated verification and 112. Volunteers are pre-seeded with Verified flags; WhatsApp, QR and voting only if time allows.

Suggested split for 4 people: hardware and firmware; backend and dispatch; volunteer app and flashcards; portal, dashboard and slides.

23. Roadmap, impact, KPIs and honest limits

SafeCircle grows in three phases, and each move forward is gated by measured results, not by a date. The gate targets below are proposals (est.) to agree as a team.



roadmap · 3 phases, 2 gates

Gate 1 is about the system working; Gate 2 is about real people responding in time, plus the legal and 112 permissions that city scale needs.

23.1 What is working in the demo vs Roadmap

Working in the demo: button with hold, cancel and status patterns; unified backend; registry and profiles; tiered alerts with the 30 s upgrade (demo speed); volunteer web app with accept, status and flashcards; family status page; dashboard; simulated verification call and 112 handoff; WhatsApp "HELP" if time allows; GPS location from the Go pendant prototype over the hotspot.

Roadmap, never shown as working: native app with wake-up location; real verification calls; DigiLocker checks; real ERSS-112 integration; the pendant's own 4G link; LoRa; fall detection; voice prompts; masked calling.

23.2 Impact

- Reaches people who cannot use an app.
- Neighbours respond in the minutes before official help; the Stockholm trial showed geofenced volunteer alerts raised bystander CPR before the ambulance from 48% to 62% (Section 3.3).
- Families get live status instead of silence.
- Authorities get one complete incident instead of fragments.
- A forced cancel no longer means help stops.

23.3 KPIs to measure in the pilot

KPI	Target (est.)	Why
Press to first volunteer alert (median)	Under 10 s	The system must never be the slow part
First accept within 60 s	80% of drills	Gate 2
Responder on scene before the ambulance	Track and report	The core promise
Verified volunteers within 1 km of each registered person	At least 3 for 90% of people	Coverage gaps show where to recruit
False alarms caught by cancel verification	Track	Tunes the countdown option and press timings
Verification calls answered	Track	Shows whether call-based checks suit seniors
Volunteers reporting noticeable battery drain	Under 10%	Tests the Section 11 design
Buttons silent for over 24 h	Under 2% of devices	Heartbeat health
Vote rejection rates by group	No unexplained gaps	Fairness check (Section 12.3)
Location accuracy at dispatch, outdoors (median)	Under 30 m	Volunteers find the person without searching
Press to first GPS fix (median)	Under 30 s	Tests the backup battery and assisted start
Days between Go pendant charges	3-7 days	Tests the power design in Section 18.5

23.4 Honest limits (say these ourselves)

- 112 integration and the verification call are simulated in the demo.
- Volunteer coverage will be thin at night and in new areas; the guard, family and 112 remain the backbone.
- Wi-Fi range, button battery life and false alarms are real limits, each with a stated mitigation. GPS fails indoors and in basements (the Wi-Fi estimate covers many buildings), and the Go pendant needs a SIM plan and regular charging.
- iPhone wake-up location and Critical Alerts need Apple approval that may be refused.
- India's Good Samaritan law covers road accidents only (Section 20.4).

- Needs testing with real users, NGOs and disability groups; we do not claim to solve every disability.

24. Risks and failure modes

The biggest risk is not technical: it is that nobody nearby responds, or that someone trusted misuses the system. Every risk below has a mitigation that already exists in this design; the last column says where.

Risk	What goes wrong	Mitigation	Where
No responder nearby	Thin coverage at night or in new areas	30 s upgrade, wider radius, guard and family calls, 112 at 90 s; coverage KPI drives recruiting	8.4, 9.2, 23.3
False alarms	Volunteers stop trusting alerts	2 s hold, verification on cancel, drills, caregiver coaching, optional countdown	6, 5.1
Volunteer misconduct	A bad actor reaches a vulnerable person	ID check, community vote, address only after accepting, verified-only for protected profiles, audit log, removal	12, 19.4
Volunteer gets hurt	Violent scene, road risk	"Observe and call" rule, safety module, "I'm in danger" button, walking-first culture	13.1, 10.1
Button fails silently	Dead battery, no Wi-Fi	Heartbeat, retry with red pattern; GSM or LoRa on the roadmap	7, 18.7
Phone kills the alert	Battery savers, silent mode, Apple entitlement refused	Setup checklist, test alert, SMS and voice fallback, time-sensitive notifications	10.3
Provider dependency	WhatsApp pricing or policy change, telephony KYC delays	Several channels for every message; start provider KYC early	4, 15.1

Risk	What goes wrong	Mitigation	Where
Legal liability	Volunteer or platform blamed for an outcome	Code of conduct, mission log, insurance, legal review, honest terms	20.4, 20.5
Data breach	Medical and address data leak	Minimal fields, field encryption, retention jobs, access logs	5.2, 19.4, 20.2
Biased vouching	Tenants, workers or minorities kept out	Reasons required, appeals, verifier path, monitored rates	12.3
Over-reliance	Families wait for SafeCircle instead of calling 112	"Adds to 112" messaging everywhere; P1 tells 112 at once	8.2, 20.5
Alert fatigue	Volunteers mute the app	Caps per day, quiet hours, fair rotation, decline reasons tracked	9.2
Mass emergency	Flood or cyclone floods the queue	Mass-incident mode with dashboard triage	16.5
Wrong first-aid advice	A card is outdated or misread	Recognised sources, doctor sign-off, review dates, no doses	13.3, 14.4
No funding after the pilot	Buttons and running costs unpaid	Open question: RWA subscription, CSR, municipal or police partnership	26
No GPS fix	Indoors, basements, lifts	Wi-Fi scan, known places, cell, bystander QR; Bluetooth finder (Roadmap)	18.2, 16.7
Pendant battery flat	GPS and 4G drain faster than Wi-Fi	GPS off between alerts, charging dock, 20% warning to the caregiver	18.5
Location misuse	Someone uses the pendant to track an adult	Location only during alerts and drills by default; adult consent for any other mode; access logged	20.6
No mobile signal	Rural roads, lifts	Store and retry, saved Wi-Fi if near, first message sent when signal returns	18.4

25. Judge Q&A and deck outline

25.1 Anticipated questions

Question	Answer
Why not just use 112?	We extend it. We cover the minutes before official help, hand off to 112, and our triggers work without a phone. 112 already lists IoT and external signals as channels, so integration is a realistic next step.
Is there evidence volunteers arrive in time?	Yes. In Stockholm, alerting CPR-trained volunteers within 500 m raised bystander CPR before the ambulance from 48% to 62%. GoodSAM runs the same model with UK ambulance services.
What if an attacker forces a cancel?	A cancel triggers a verification call. No answer, a wrong PIN or the duress PIN keeps the alert alive at high priority, while the button stays quiet.
Accidental presses?	2 s hold to trigger, longer hold to cancel, verification, monthly drills, optional countdown.
How does the person know help is coming?	Light, sound and vibration for each stage: sending, received, accepted, arriving, arrived. Works for blind and deaf users.
Won't "I am available" drain volunteers' batteries?	No constant GPS. A coarse area refreshed about hourly or on movement, one exact reading only when an alert is near them, live GPS only after they accept, stopping on arrival.
Why is the nearest person not always first?	Tier 1 is the 3 nearest verified volunteers, with a small bonus for matching skills such as a nurse for a medical case. Everyone else is alerted at the same time and upgraded at 30 s.

Question	Answer
Can volunteers be trusted?	Government ID through DigiLocker, approval by neighbours who know them or a society verifier, address only after accepting, audit logs, removal for misconduct.
Isn't a neighbourhood vote biased?	It can be. Concerns need reasons, rejections can be appealed, a verifier path stays open and rejection rates are monitored.
What do volunteers do on arrival?	Smart flashcards predict the 3 likely situations from the profile and history; they tap what they see and follow the card. CPR is always one tap away.
What if nobody responds?	Guard and family are alerted from second zero; 112 is told at 90 s, or at once for critical cases.
Privacy?	Minimal fields, staged disclosure enforced on the server, coarse volunteer location, short retention, DPDP-ready consent including parents and guardians.
Are volunteers legally protected?	Only partly. India's Good Samaritan section covers road accidents. We use a code of conduct, insurance and legal review, and we say this openly.
How does someone without the button raise an alarm?	WhatsApp "HELP" from a registered number, a caregiver message, or a stranger scanning the pendant QR.
Is the 112 link real?	Simulated in the demo; real integration needs state permission.
What does it cost?	Prototype about ₹5,500-15,000; pilot about ₹3-4.5 lakh plus monthly running costs for 50-100 homes (est.).
How do you find someone who is not at home?	The Go pendant has GPS and 4G. It sends the best location it has within seconds (a saved Wi-Fi place, a Wi-Fi scan or the cell tower), refines it with a GPS fix, and keeps updating while the alert is open.

Question	Answer
Isn't the pendant a tracker?	Not by default. It shares location only during an alert or a drill; any other mode needs the adult's own consent, or a parent's for a child.
Why not GPS in every button?	A wall or bedside button never moves, so the registered address is exact and the battery lasts far longer. GPS and 4G go only in the pendant people wear outside.
What if a beep would put a girl in more danger?	Stealth mode: the button confirms each stage with light only, with no sound, no vibration and no calls to her phone, so people nearby cannot tell help is coming.

25.2 Deck outline (12-14 slides)

1. Title and one-line pitch (Section 1)
2. The broken chain and the hidden requirement (1)
3. Who we serve: four user groups (17)
4. Existing systems, prior art and gaps (3)
5. Two faces, one backend (2)
6. Face A: triggers and setup portal (4, 5)
7. Press, cancel and verification call (6)
8. Status the person can feel (7)
9. Tiered alerting and ranking (9, 8)
10. Face B: the volunteer app and battery design (10, 11)
11. Trust, training and smart flashcards (12, 13, 14)
12. Architecture, hardware and GPS location (18, 19)
13. Demo flow (22)
14. Roadmap, KPIs, cost and honest limits (22, 23)

For a 7-slide version: merge 2-3, 4-5, 6-8, 9-11, 12-13, keep 1 and 14.

Style from v1: deep navy, teal and amber; Cambria headings, Calibri body. The cyclone deck was a style reference only.

Placeholders still to fill: final project name, team name and members, team ID, photos of the prototype and dashboard.

26. Open questions for the team

Every "Decision needed" in this document is listed here. Tick each one off once the team agrees, and update the section it points to.

26.1 Product decisions

- [ ] Send instantly on a 2 s hold, or keep a 3 s countdown before sending? (6.1)
- [ ] After a cancel that cannot be verified, should the button stay quiet (safer against an attacker) or show "help still coming" (more reassuring)? (6.2)
- [ ] Use a duress PIN on the verification call, or only "press 1 safe, press 9 help"? (6.2)
- [ ] Unregistered "HELP" messages: guard desk only, or volunteers too? (4, T2)
- [ ] Responders per case type: up to 3 for collapse, 1 plus backup for a fall, and so on? (8.5)
- [ ] Timers: 30 s upgrade and 112 at 90 s? (8.4)
- [ ] Tier 1 size (3) and radius steps (1.5, 3, 5 km)? (9.1, 9.2)
- [ ] Can an unverified volunteer ever be the sole responder for a child or a person with a cognitive disability? (9.2)
- [ ] Community vote pass rule: at least 3 vouches, two thirds of non-abstain votes, within 7 days? (12.3)
- [ ] Retention periods for location fixes, chats and incident records? (11.4, 20.2)
- [ ] Stealth mode: lights only as specified, or also a silent vibration only she can feel? How dim should the light be, and where on the pendant should it sit? (7.2)

26.2 Build and demo

- [ ] Demo format and deadline: live on stage, or video plus slides?
- [ ] Team size and role split (suggested split in 22.4)
- [ ] Stack: Node or Python; React or plain HTML; Flutter or React Native for the pilot app
- [ ] Which two personas to demo (for example a senior and a non-verbal child)?
- [ ] Family and volunteer alerts in the demo: WhatsApp templates, Telegram, or app only?
- [ ] Final project name and team details

26.3 Pilot and partners

- [ ] Which city, neighbourhood or society hosts the pilot?
- [ ] Doctor or medical college to review tutorials and flashcards (13.3)
- [ ] Lawyer to review ID checks, consent and volunteer liability (20)
- [ ] Partner for hands-on CPR workshops (Red Cross, hospital, NGO)
- [ ] Local police or PSAP contact for a future 112 link

- [ ] Funding after the pilot: RWA subscription, CSR, municipal or police partnership?

26.4 GPS and the Go pendant (v2.1)

- [ ] Pilot pendant route: LilyGO T-A7670G with GPS, or our own board? (18.6)
- [ ] Demo GPS module: budget NEO-6M or ATGM336H, or low-power MAX-M10S? (18.6)
- [ ] Do we need NavIC support? (18.3)
- [ ] IoT SIM provider and data plan (18.4)
- [ ] Location update interval while moving and while still (18.2)
- [ ] Raise to P1 when a child's pendant moves at vehicle speed? (8.2, 16.7)
- [ ] Any location mode beyond alerts (safe-zone), and who may switch it on (5.1, 20.6)
- [ ] Charging target and dock design (18.5)

27. Sources and caveats

Sources consulted for v2 on 1 Oct 2026, grouped by topic. v1 sources are listed separately and were not re-checked.

27.1 India: 112 and ERSS

- MHA: Emergency Response Support System (10 channels, including IoT and external signals)
- MHA: Nirbhaya/ERSS section (ERSS 2.0 in 24 states and UTs)
- Rajya Sabha answer, 20 Dec 2023 (ERSS in all 36 states and UTs)
- 112.gov.in: About ERSS
- OneIndia: SHOUT volunteers

27.2 Prior art and evidence

- GoodSAM FAQ
- NIAS GoodSAM public FAQ (500 m, 15 s, battery slider, code of conduct)
- St John NZ: GoodSAM and GoodSAM poster
- Karolinska Institutet: SMS Lifesavers and CBS News on the trial

27.3 Platform rules

- Meta: WhatsApp Business Platform pricing
- Android: foreground service restrictions
- Google Play: full-screen intent policy
- Apple: Location Push Service Extension

- Apple forums: location push on locked devices
- Apple forums: Critical Alerts eligibility

27.4 Providers

- Exotel Voice v1 API and Exotel services
- Home Assistant: ESP Web Tools and Improv Wi-Fi
- Cashfree: DigiLocker FAQ

27.5 Law, data protection and health

- SCC Online: DPDP Rules 2025 highlights
- AMLEGALS: DPDP Rules timeline
- Supreme Court order, 26 May 2026 (history of Section 134A)
- Daily Star: Section 134A explainer
- Guide to the NMC Indian Medical Register
- American Heart Association: CPR guideline FAQ

27.6 v1 sources (not re-checked)

Vikaspedia (ERSS and Kavalan pages), Business Standard (112 rollout, 2019), Andaman Sheekha, PIB and MHA summary via journalsofindia.com, Anantamias explainer, The News Minute (Kavalan launch), DT Next (Kaaval Karangal), Meta and 360dialog WhatsApp docs, Twilio support notes on India SMS, MSG91 and 8x8 inbound SMS docs.

27.7 Caveats

- Some sources are secondary (explainers, vendor docs, developer forums). Re-check anything that becomes a slide claim.
- The AHA page reflects the 2015 guideline cycle; confirm against the newest AHA, ILCOR or Indian Resuscitation Council guidance.
- Platform rules (Android, Apple, WhatsApp) change often; re-check before building.
- Prices, timings, targets and component specs marked (est.) are estimates.
- Nothing here is legal advice; get a qualified review of ID checks, consent and volunteer liability.
- No real 112 or ERSS integration exists in this project. Any handoff shown is a simulation.
- Phone-maker SOS features and paid family-safety apps were not researched.

27.8 GPS, cellular and positioning (v2.1)

- LilyGO T-A7670G and T-A7670 series FAQ (bands, GPS version, data-only)

- Probots: T-A7670G R2 with GPS (Indian price)
- SparkFun MAX-M10S and Tanotis listing (start times, power, Indian price)
- SparkFun SAM-M10Q (accuracy, aided start, current)
- Hubtronics: Quectel L89HA-S90 (NavIC module)
- Telit: NavIC and AIS-140
- Google Geolocation API and request and response examples
- ESPWifiLocation library (ESP32 Wi-Fi scan to the Geolocation API)

Module prices are retail listings seen on 1 Oct 2026. GPS accuracy and start times are datasheet figures; test in real streets, buildings and pockets before quoting them.